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Reconfiguration Decision Analytics for Evacuation Traffic and Land-Use Transition in Disaster-Prone Areas

災害リスク地域における避難と土地利用の最適遷移

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Urban transportation and land use planning adaptable to disaster.

Short term: Evacuation

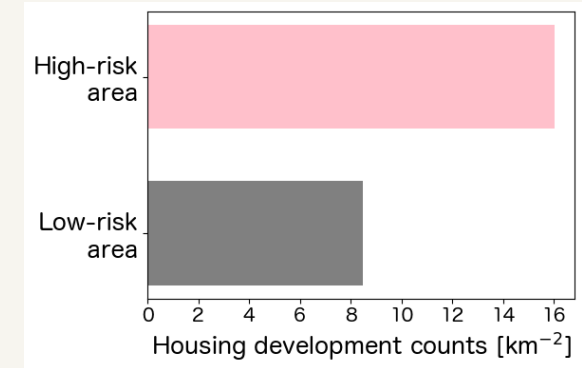
Heavy **traffic congestion** during evacuation



▶ **Demand-supply imbalance during evacuation** must be solved.

Long term: Land use

Damage to infrastructure and assets in high-risk areas



Housing supply in Matsuyama (2008-2023)

▶ **Controlling population in high-risk areas** is needed.

Planning of “sequential” and “staged” transition process of evacuation and land-use

↔ static or one-shot planning

- “Sequential” for **reachability** between the current and desired state.
- “Staged” for **flexibility** and **adjustability** under the uncertainty of disaster occurrence.

1. Reconfiguration decision analytics as sequential decision analytics
2. Reconfiguration analysis of evacuation traffic control and land-use in disaster-prone areas.

Chapter 1,2

Reconfiguration decision analytics

1. Fixed terminal state

$$S_T = Z^S \text{ final state}$$

2. Fixed initial and terminal decision

$$x_{t+1} = \mathcal{F}(x_t) \text{ adjacency relation}$$

$$x_0 = A^x, x_T = Z^x$$

Chapter 5 Dynamic game of land-use in disaster-prone areas

Chapter 4

Information design of evacuation traffic control

Interaction
 $\theta^{1,2}, \theta^{2,1}$

Model
 S^M

- Land-use and transport supply
- Evacuation planning

Chapter 3

Method of sampling-space restriction

$$\max_{\pi} \mathbb{E} \left\{ \sum_{t=0}^T \rho^t C(S_t, X^{\pi}(S_t)) \mid A \right\}$$

$$\text{s.t. } S_{t+1} = S^M(S_t, x_t, W_{t+1})$$

$$x_t = X^{\pi}(S_t) \in \mathcal{X}_t, S_0 = A^S$$

Solution

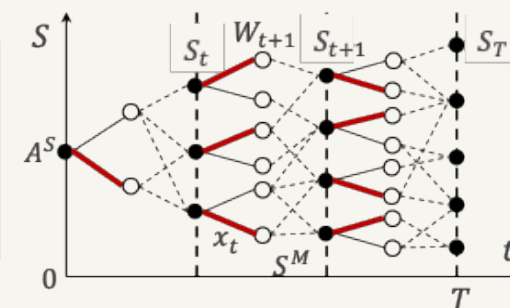
Policy π^*

Information

Dynamic expectations of multiple players

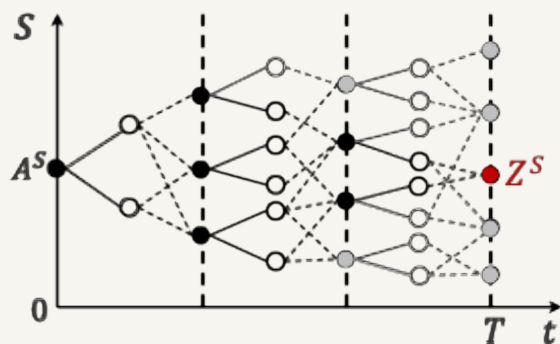
Short- and long-term disaster risk reduction through traffic control, information design, and location incentives.

$$\max_{\pi} \mathbb{E} \left\{ \sum_{t=0}^T \rho^t C(S_t, X^{\pi}(S_t)) \mid A^S \right\} \quad \text{s.t.} \quad \begin{aligned} S_{t+1} &= S^M(S_t, x_t, W_{t+1}) \quad (\text{transition function}) \\ x_t &= X^{\pi}(S_t) \in \mathcal{X}_t \quad (\text{decision feasibility}) \\ S_0 &= A^S \quad (\text{initial state}) \end{aligned}$$



1. Fixed-terminal-state problem

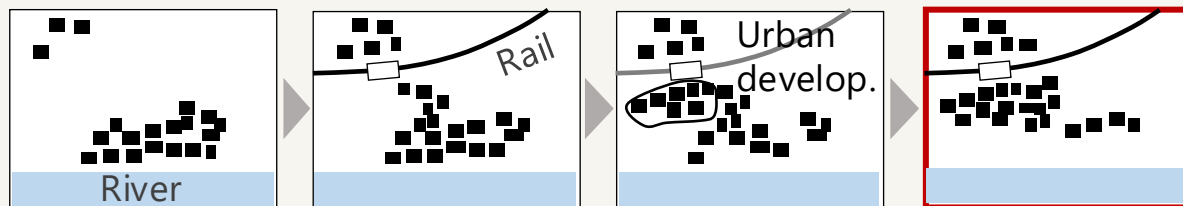
Base problem + constraint $S_T = Z^S$



Single-agent problems:
Maximum principle

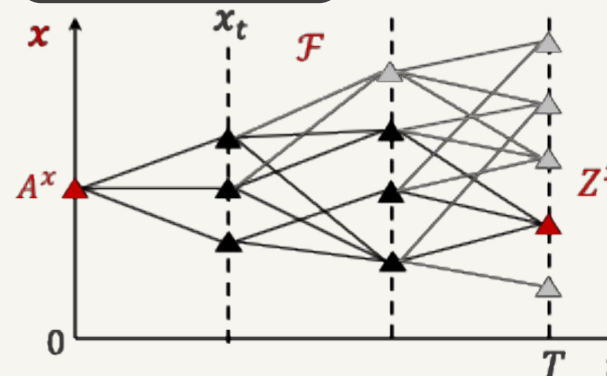
$$\begin{aligned} \max_u H \\ \dot{S} &= \partial H / \partial \lambda \\ \dot{\lambda} &= -\partial H / \partial S \\ S(T) &= Z^S \end{aligned}$$

e.g., Land-use transition = **Multiple agents** Z^S



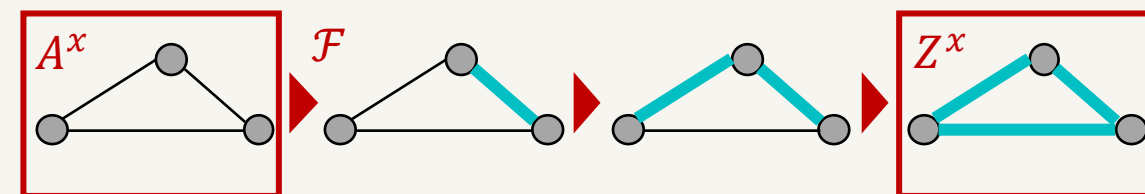
2. Fixed-initial-and-terminal-decision problem

Base problem + constraints $x_{t+1} = \mathcal{F}(x_t)$ adjacency relation
 $x_0 = A^x, x_T = Z^x$



Continuous decisions : Model
Predictive Control

e.g., Staged deployment of traffic control = **Discrete**



2. Fixed-initial-and-terminal-decision problem

Base problem

+constraints

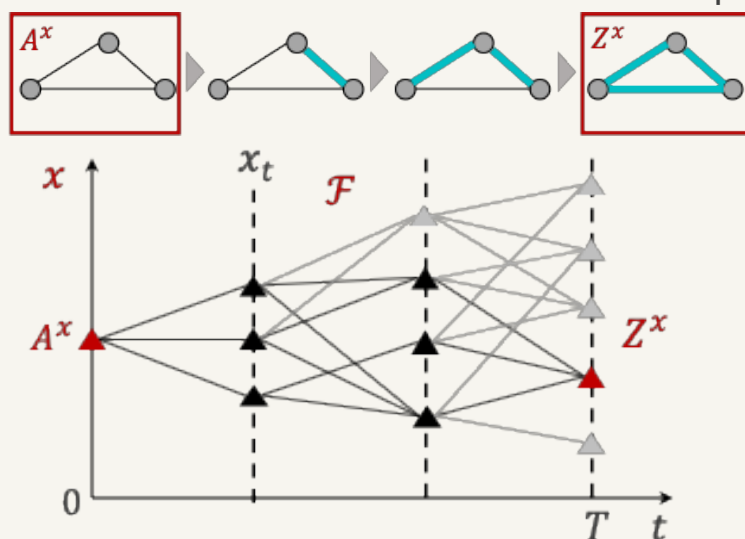
$$x_{t+1} = \mathcal{F}(x_t) \text{ adjacency relation}$$

$$x_0 = A^x, x_T = Z^x$$

Discrete decisions : **Combinatorial reconfiguration** (Nishimura, 2018)

- All transition paths: exponentially large. → Direct methods using **sampling algorithm**
- Feasible paths: many, but much smaller than all paths. → Sampling inefficiency

e.g., Choose at most one out of r alternatives per timestep



$T = 7$

r	Unconstrained paths	Feasible paths	Sampling efficiency (%)
5	15625	2763	17.68
10	1000000	11281	1.12
15	11390625	11322	0.10
20	64000000	8877	0.013
30	729000000	7060	0.00096

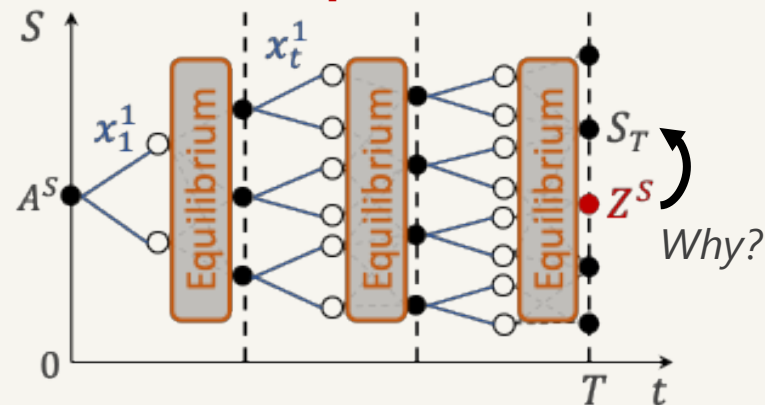
Random-sampling-based algorithms are inefficient. → Need of efficient sampling algorithm.

1. Fixed-terminal-state problem

Base problem + constraint $S_T = Z^S$

Multiple-agent interaction

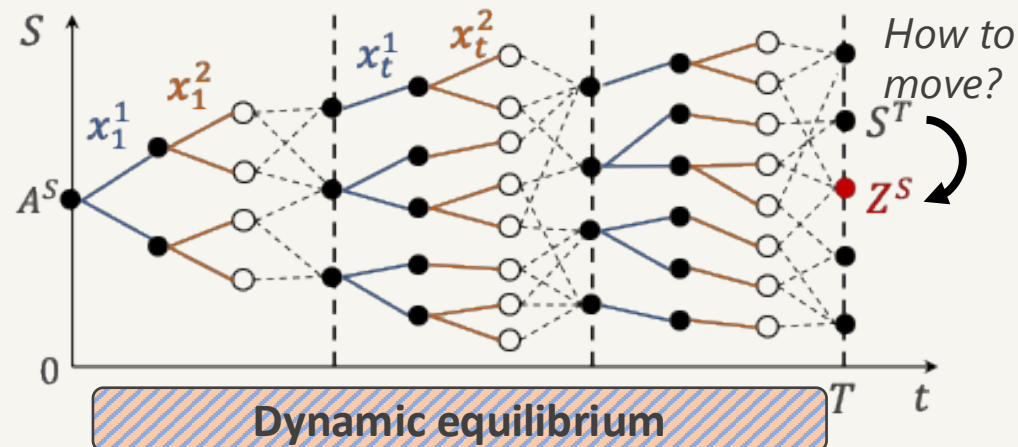
- **Information paradox** (Arnott et al., 1991)



Announcing player 1's decisions ex ante **shifts the equilibrium in player 2**, leading to suboptimal ex post outcomes.

$$S^M(S_t, x_t, W_{t+1}) \rightarrow S^M(S_t, x_t, W_{t+1}, \underbrace{x_{t+1 \sim T}}_{\text{information}})$$

- Learning transition models S^M **with dynamic expectations** and control.



Player 1 optimizes their decisions $\{x_1^1, x_2^1, \dots, x_T^1\}$, anticipating the **best response of players 2** $\{x_1^2, x_2^2, \dots, x_T^2\}$.

→ **Learning** interaction **to transition the system** toward the desired one **using the interaction**.

Chapter 3

Reconfiguration optimization for contraflow planning of urban evacuation networks

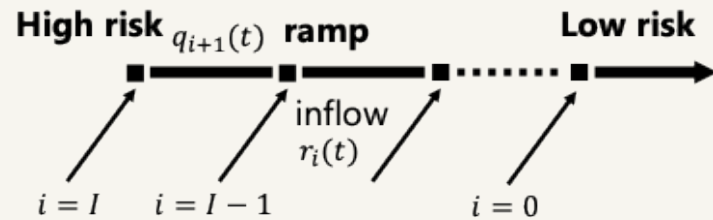
In the future, the search for adaptive strategies should be extended to more complex, two-dimensional transportation systems. ... an evacuation strategy of this type should take into account that people can change routes and may not utilize the network efficiently.

So, Stella K., and Carlos F. Daganzo.

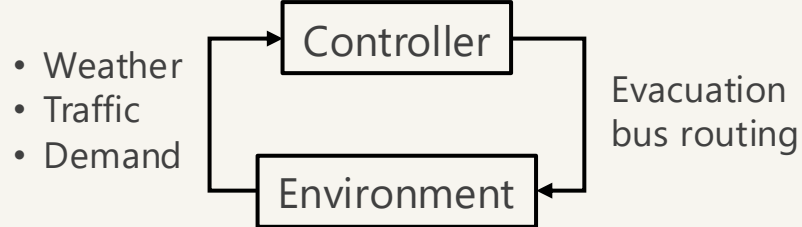
"Managing evacuation routes." *Transportation research part B: methodological* 44.4 (2010): 514-520.

Online control, e.g., Ramp metering, routing

So & Daganzo (2010): InFO control policy



Cui+ (2025): Reinforcement learning

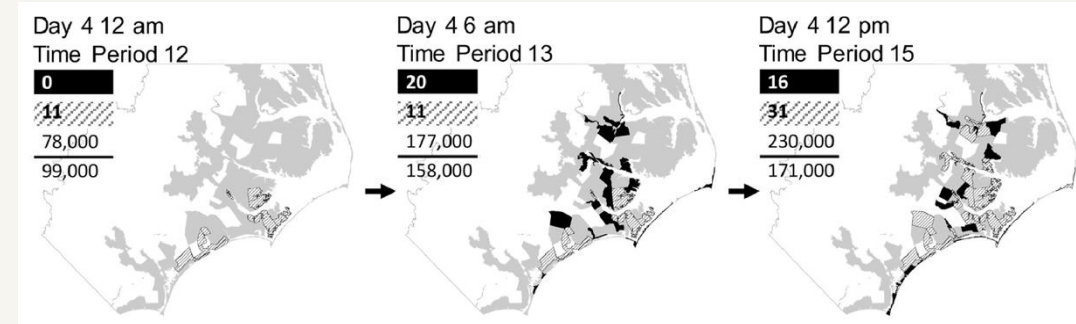


- × **High control cost** ← reactive
- × **Implementation challenges**

Proactive strategic planning of traffic control

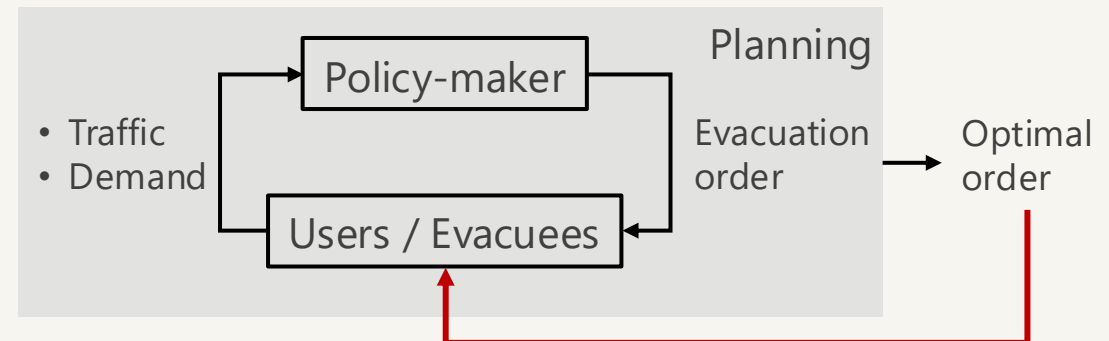
Offline planning, e.g., Shelter location allocation, staged evacuation

Yi+ (2017): Staged evacuation order



Enumerate all transition paths and evaluate by DTA

→ **Impossible as transition paths become large.**

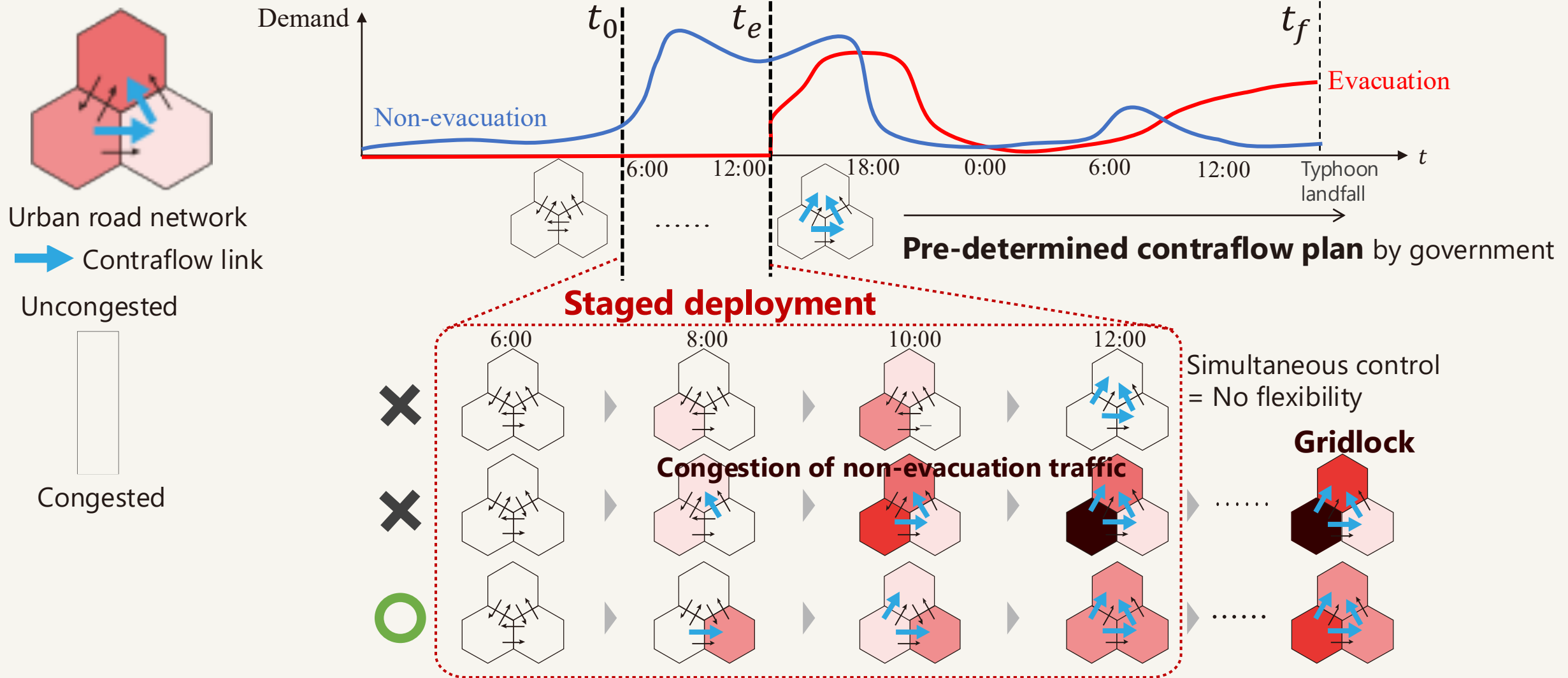


Users interactively respond to the planned policy.

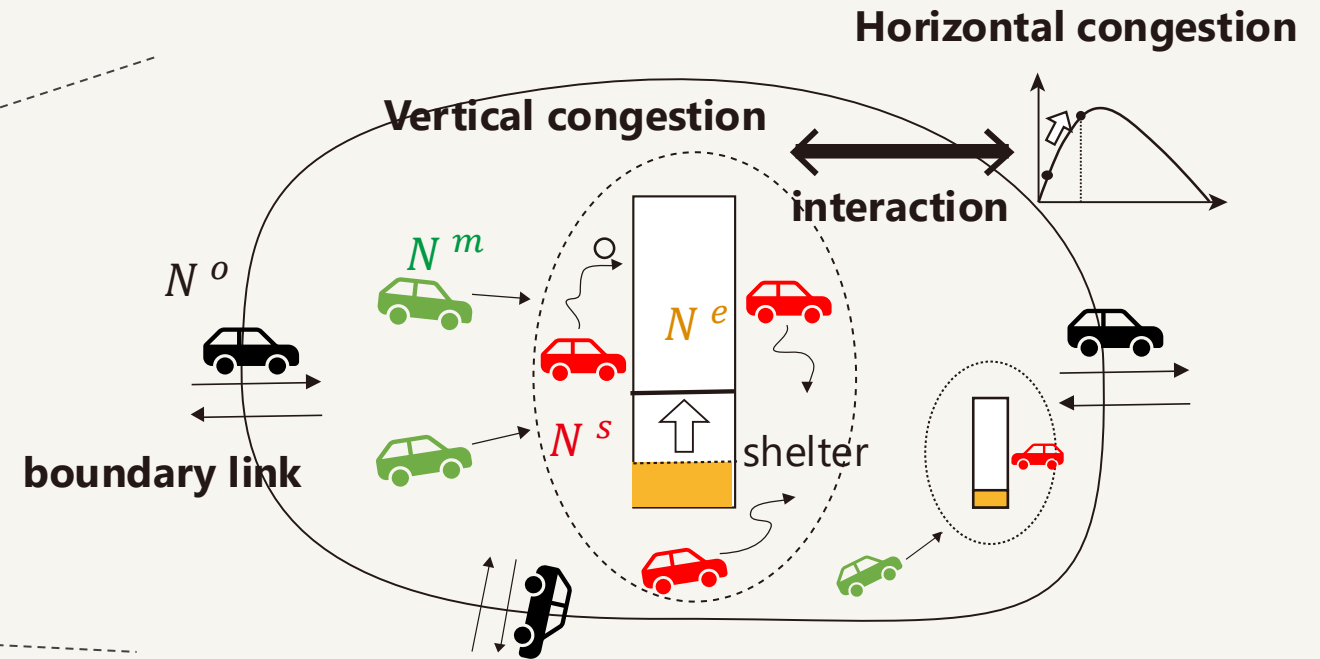
→ **Information paradox (time inconsistency)**

Reconfiguration of disaster traffic control

9



Optimal reconfiguration from normal to disaster traffic control with efficient algorithm



Evacuation traffic

N^o : Moving to other zones

N^m : Moving within destination zones

N^s : Searching for shelters

N^e : Completed evacuation

Non-evacuation traffic

$N^{o,back}$: Moving to other zones

$N^{m,back}$: Moving within the zone

N^p : Reaching their destination

Optimal reconfiguration problem

11

$$\max_{X_{\tau_0+1} \dots X_{\tau_e-1}} \quad \boxed{z_1} \text{ or } \boxed{z_2} \quad \begin{array}{l} \boxed{z_1}: \text{Minimize total travel time \textit{after} reconfiguration} \\ \boxed{z_2}: \text{Minimize total travel time \textit{during} reconfiguration} \end{array}$$

Decision variables: X_τ (contraflow configurations at time τ)

subject to (MFD-based traffic dynamics)

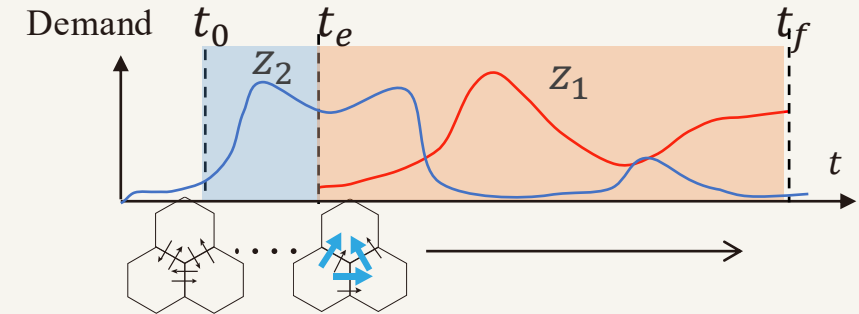
$$X_\tau \in \mathcal{F}(X_{\tau-1}), \quad \forall \tau \in \{\tau_0 + 1, \dots, \tau_e - 1\}$$

$$x_{ij}(\tau) + x_{ji}(\tau) \leq 1, \forall i \in R, j \in \Gamma_i, \forall \tau \in \{\tau_0 + 1, \dots, \tau_e - 1\}$$

$$X_{\tau_0} = S, \quad X_{\tau_e} = T$$

$$x_{ij}(\tau) \in \{0, 1\}, \forall i \in R, j \in \Gamma_i, \forall \tau \in \{\tau_0 + 1, \dots, \tau_e - 1\}$$

Graphically, the problem is to find the **optimal path of contraflow configurations** in the solution space from $\tau = t_0$ to $\tau = t_e$.

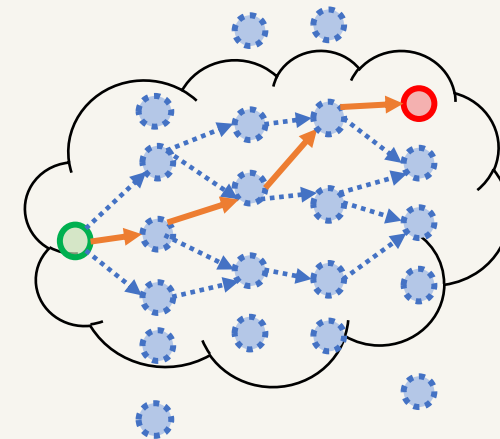


Adjacency relation $\cdots \rightarrow$

Topological constraints

Initial (●) and target (●) configurations

Binary decisions (where to activate)



Cross-Entropy Method: A sampling-based heuristics for hard optimization problems.

Sampling

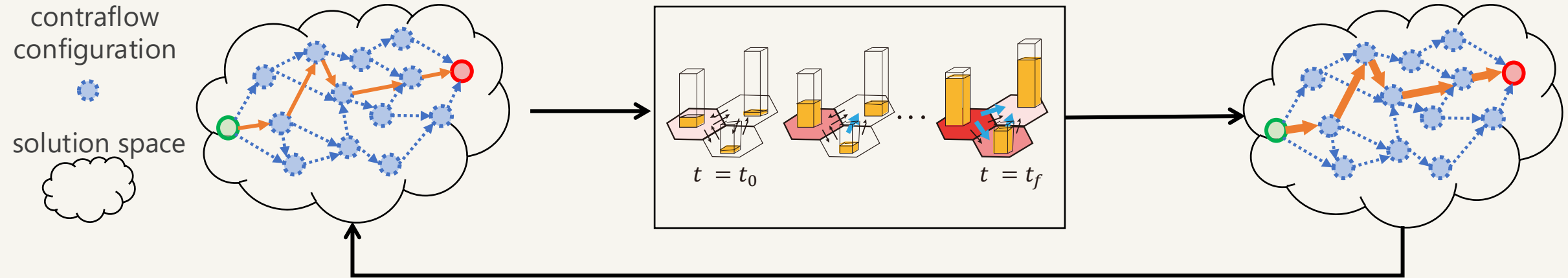
Generate reconfiguration paths from a probability distribution.

Evaluation

Evaluate the objective value of each path with the traffic simulation.

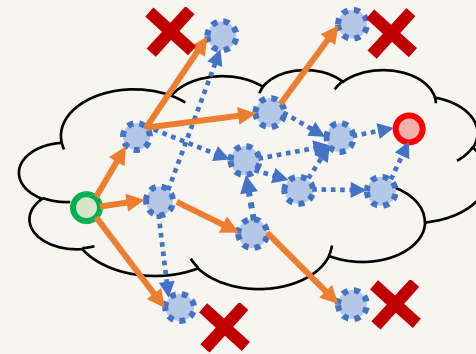
Probability update

Update the sampling probabilities based on the performance of sampled paths.



Issue:

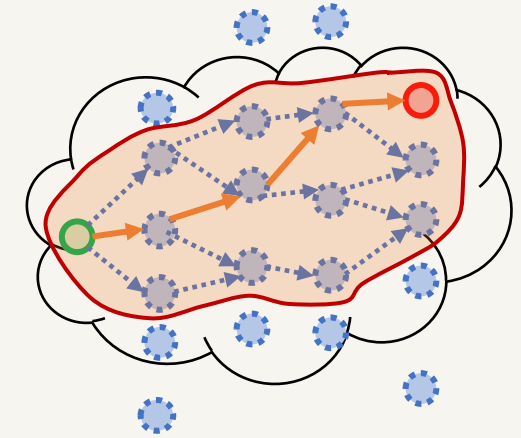
When the feasible solution space becomes small due to constraints, **the sampling efficiency deteriorates.**



Idea: If **all feasible configurations can be enumerated in advance**, then sampling can be restricted to the feasible solution space.

Requirements:

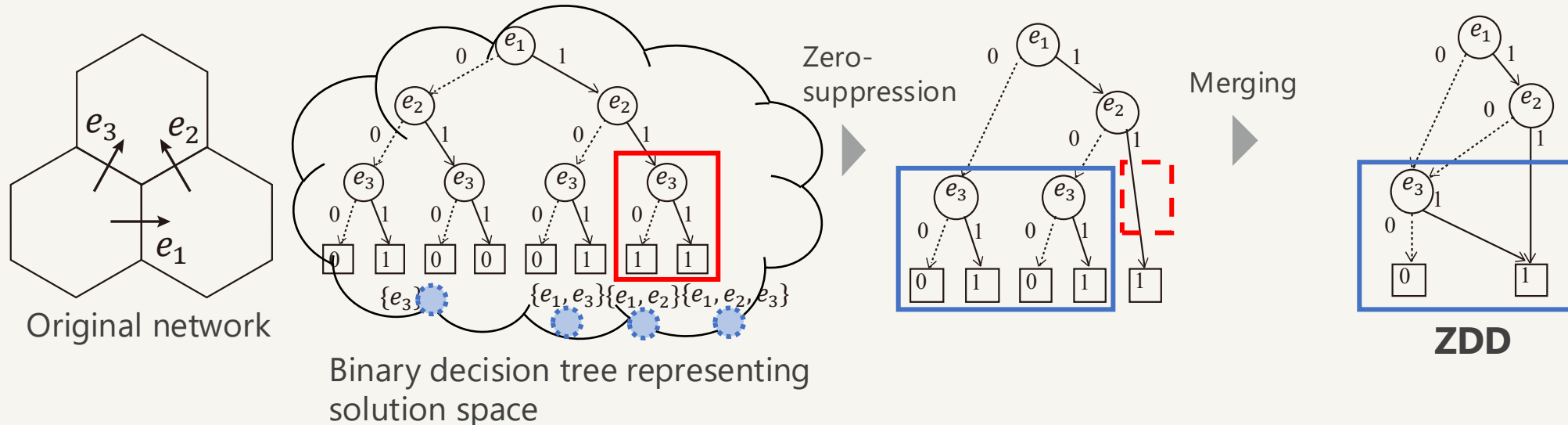
- Efficiently **enumerate and store** sets of feasible configurations.
- Perform **high-speed set operations** representing adjacency relations.



Sampling-space restriction

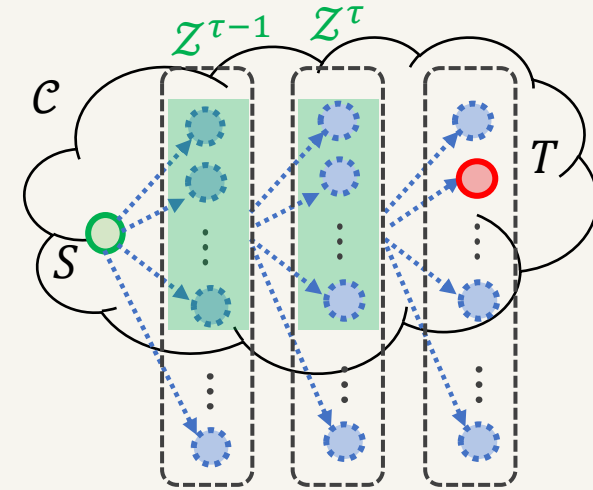
➡ Zero-suppressed binary Decision Diagram (ZDD):

A data structure that efficiently enumerates and stores sets of sets and performs fast set operations.



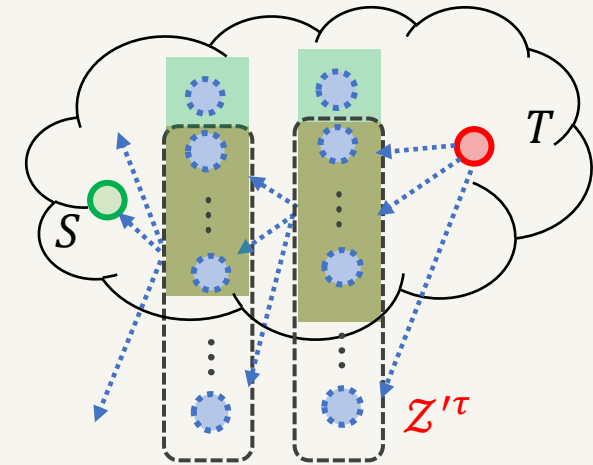
1. Forward search

1. Construct ZDD $\mathcal{C}$ satisfying the topological constraints.
2. Initialize the set of feasible configurations: $\mathcal{Z}^{\tau_0} = \{S\}$
3. Iterate from $\tau = \tau_0 + 1$ to τ_e
 - Enumerate states reachable from $\mathcal{Z}^{\tau-1}$ using ZDD operations
 - Intersect with constraint set $\mathcal{C}$ to obtain $\mathcal{Z}^{\tau}$.
 - Utilize **high-speed set operations enabled by ZDD**.
4. Check whether the target configuration T is reached.



2. Backward search

1. Start from the target configuration T and proceed **backward**.
2. At each step, enumerate states reachable from $\mathcal{Z}'^{\tau+1}$ using ZDD operations to obtain $\mathcal{Z}'^{\tau}$.
3. Intersect $\mathcal{Z}'^{\tau}$ with the forward search results $\mathcal{Z}^{\tau}$ to extract the **feasible set of transitions**.



→ Enumerate all feasible configurations within sampling space.

Comparison: Proposed sampling algorithm vs. naïve sampling.

# links	Method of sampling-space restriction		Naïve sampling
	ZDD construction [s]	Sampling [s]	
10	0.001	0.370	0.180
20	0.002	0.387	0.642
30	0.006	0.671	1.808
40	0.009	1.243	4.886
50	0.460	5.895	13.07
60	2.641	7.115	33.26

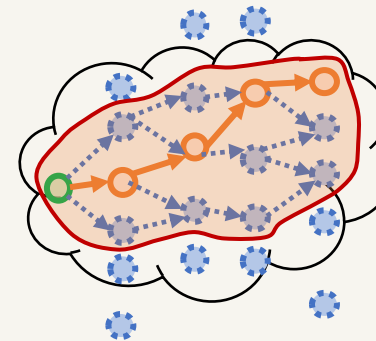
20 sec.



- ZDD sampling is **2.7–4.7 times faster** than naïve sampling.
- A 20-second difference per iteration = a **10-minute reduction in total computation time** (when repeated 30 times in the optimization algorithm)

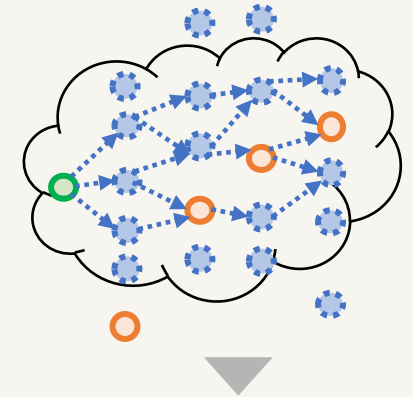
Sampling-space restriction

- Sample from the feasible space



Naïve sampling

- Sample from the whole space



- Check feasibility
- Repeat until the feasible path is found.

Comparison:

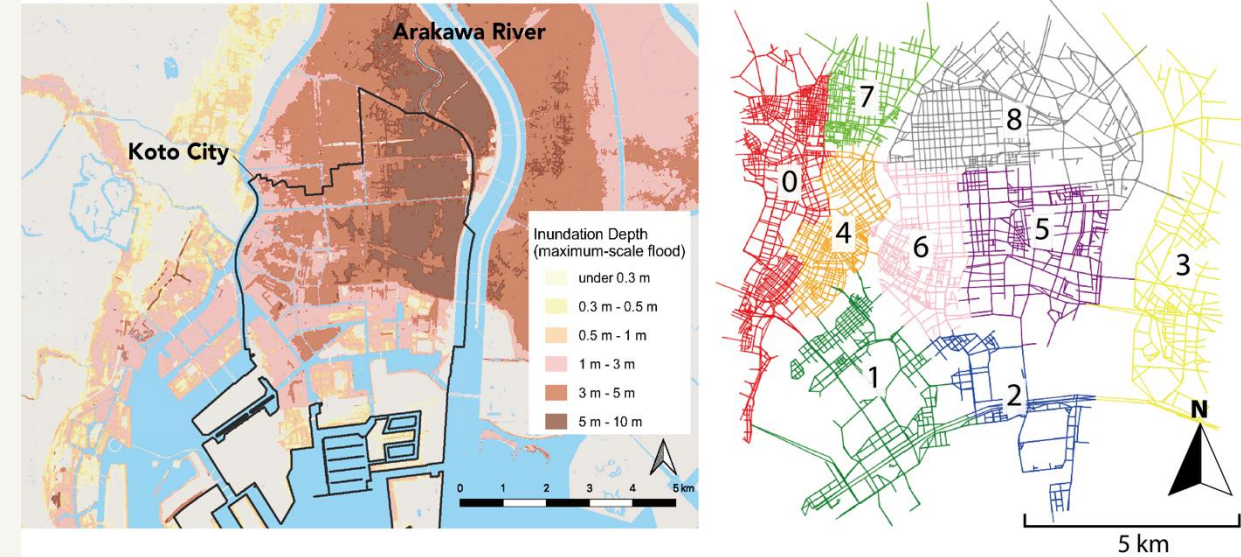
- Proposed algorithm **with ZDD**
- Proposed algorithm **without using ZDD** (BFS algorithm)

Search Depth		2	3	4	5	6	7
ZDD	CPU Time(s)	0.006	0.006	0.007	0.008	0.012	0.014
	Memory(GB)	0.240	0.244	0.248	0.235	0.243	0.243
BFS	CPU Time(s)	82.01	2095.89	NS	NS	NS	NS
	Memory(GB)	0.45	1.03	NS	NS	NS	NS
Feasible configuration		393	3,305	19,317	83,300	274,754	710,066

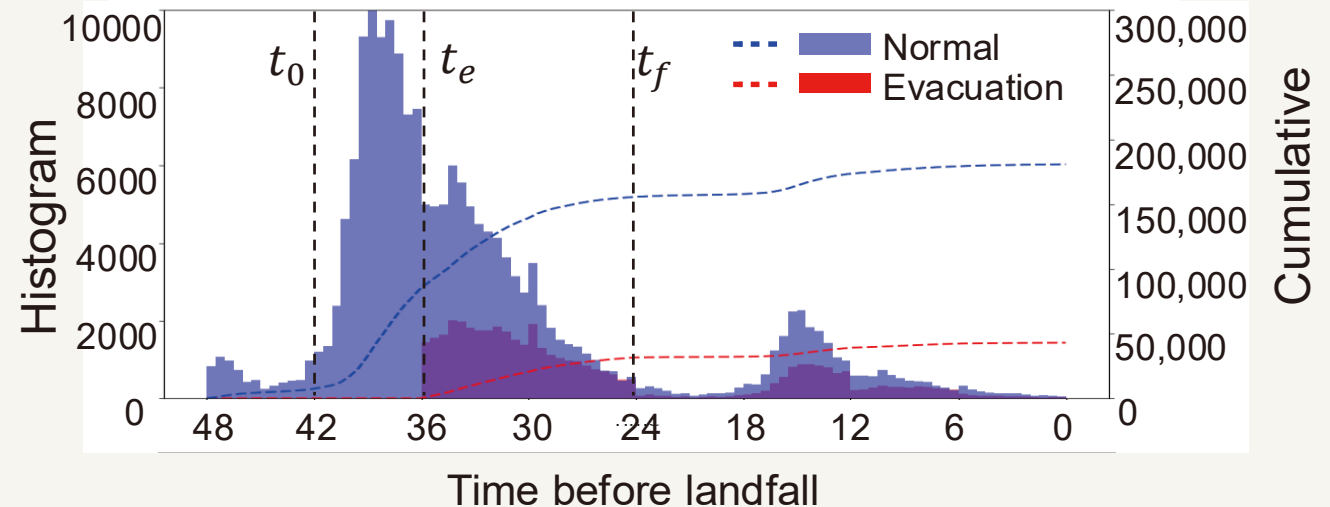
NS: computation did not complete within 1 hour

- Feasible reconfiguration paths are enumerated in about 0.01 seconds using ZDD.
- The pure BFS approach fails to scale beyond 2 steps.

- The urban network of Tokyo is divided into **9 zones** and **34 boundary links**.
- The evacuation order is issued 36 hours before landfall (t_e).
- Regular traffic peaks between 42 and 36 hours before landfall ($t_0 \leq t \leq t_e$).
- One reconfiguration step corresponds to **1 hour**.



- Normal demand:
Public survey data on road traffic
- Evacuation demand:
Generated using evacuation departure and destination choice model



Four demand scenarios that cause gridlock during evacuation periods

Max-Max

Evacuation: Max
Background: 100%

Max-90

Evacuation: Max
Background: 90%

Max-80

Evacuation: Max
Background: 80%

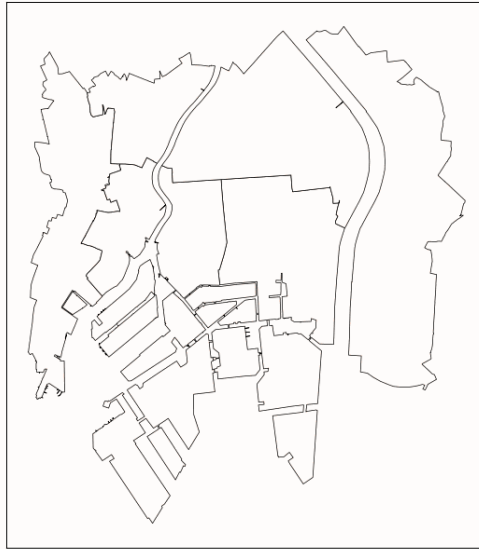
95-Max

Evacuation: 95%
Background: 100%

0.0 hour (Day 1 00:00)



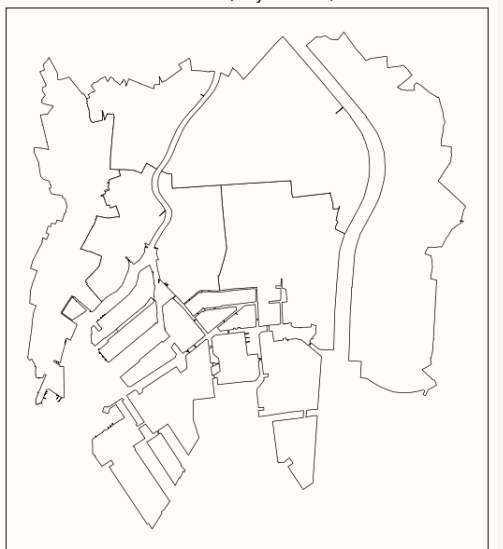
0.0 hour (Day 1 00:00)



0.0 hour (Day 1 00:00)



0.0 hour (Day 1 00:00)



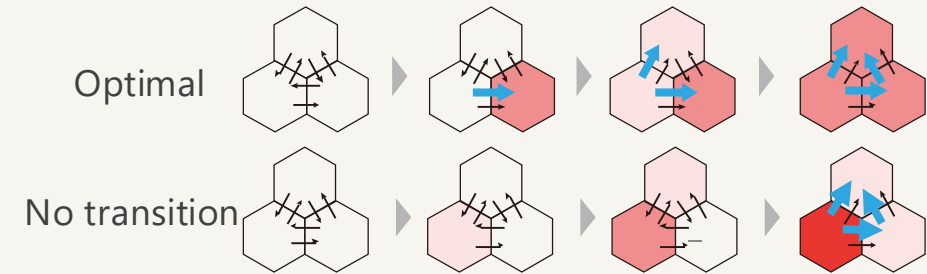
High demand scenarios

Low demand scenarios

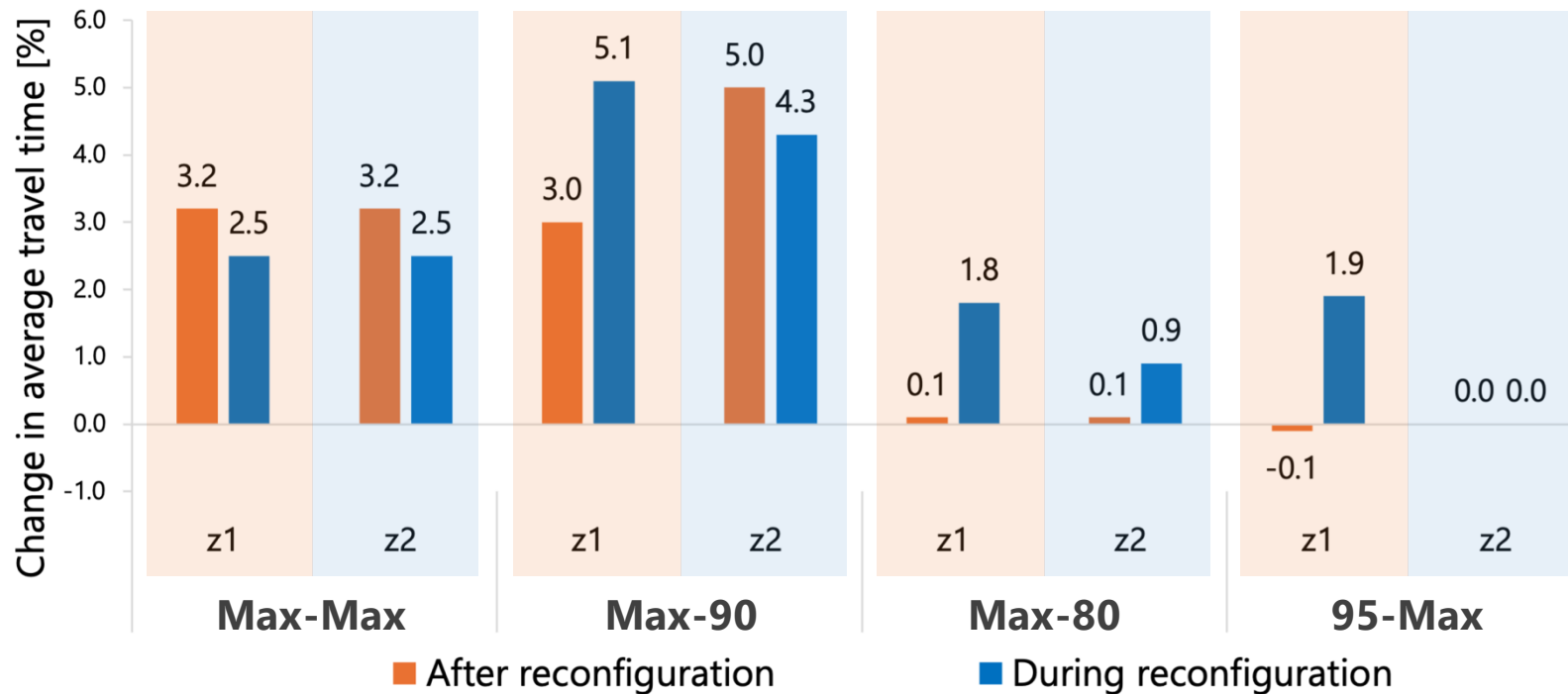
z_1 : Minimize the total travel time *after* the reconfiguration

z_2 : Minimize the total travel time *during* the reconfiguration

Evaluation metrics: Average travel time during/after the reconfiguration



Change in average travel time vs. no-transition



■ Evacuation performance $z_1 \geq z_2$

■ Performance in normal time $z_1 \leq z_2$

Reconfiguration is **worse** than the case without reconfiguration.

► Reconfiguration overall act as a disruption.

The negative impact is not so large (5.1% $\approx$ 5 min.)

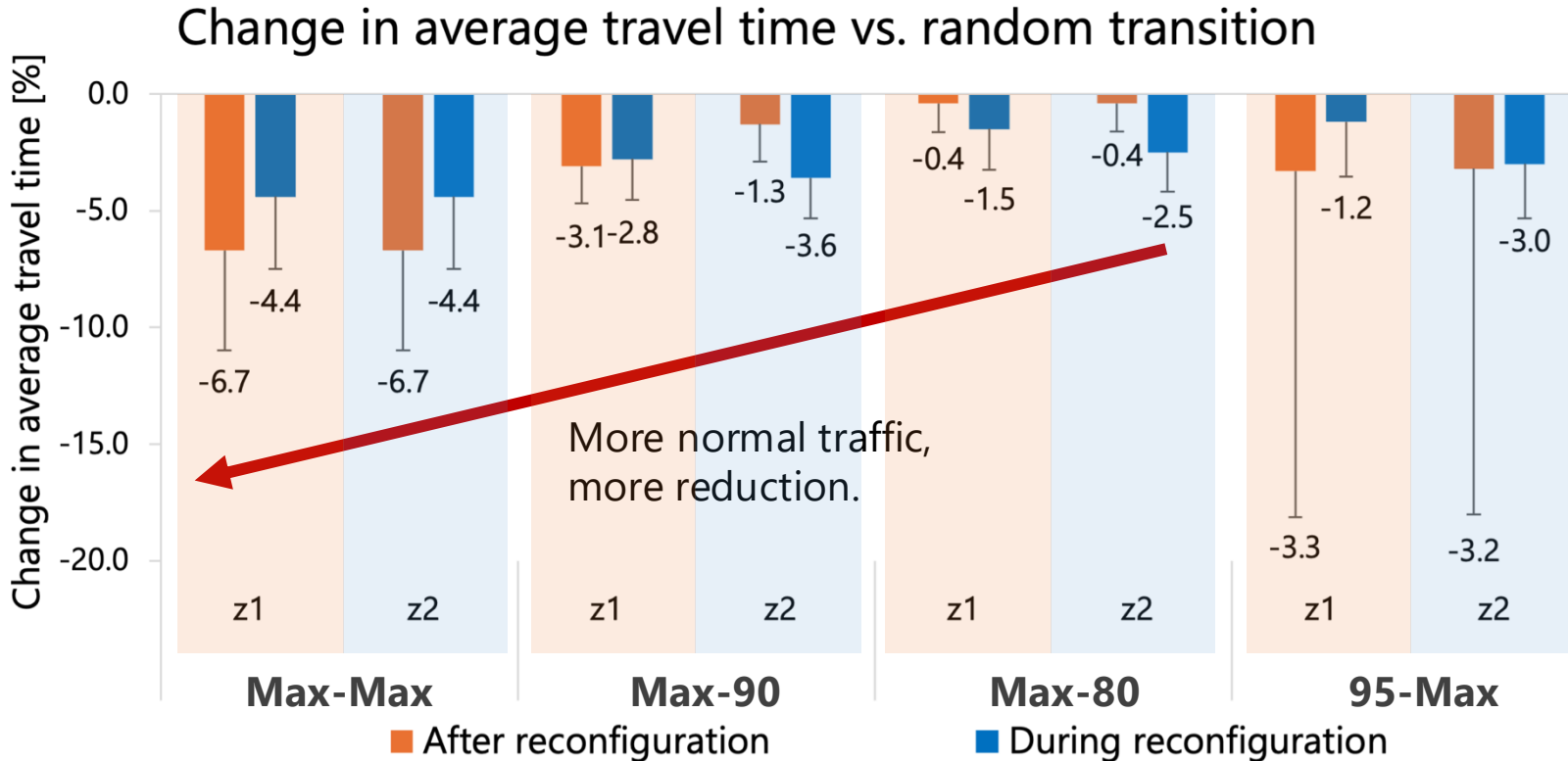
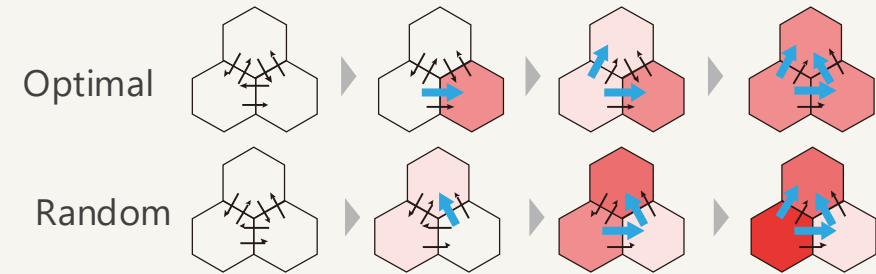
Optimal transition vs. Random transition

20

z_1 : Minimize the total travel time *after* the reconfiguration

z_2 : Minimize the total travel time *during* the reconfiguration

Evaluation metrics: Average travel time during/after the reconfiguration



The optimal transition improves traffic compared with a random one.

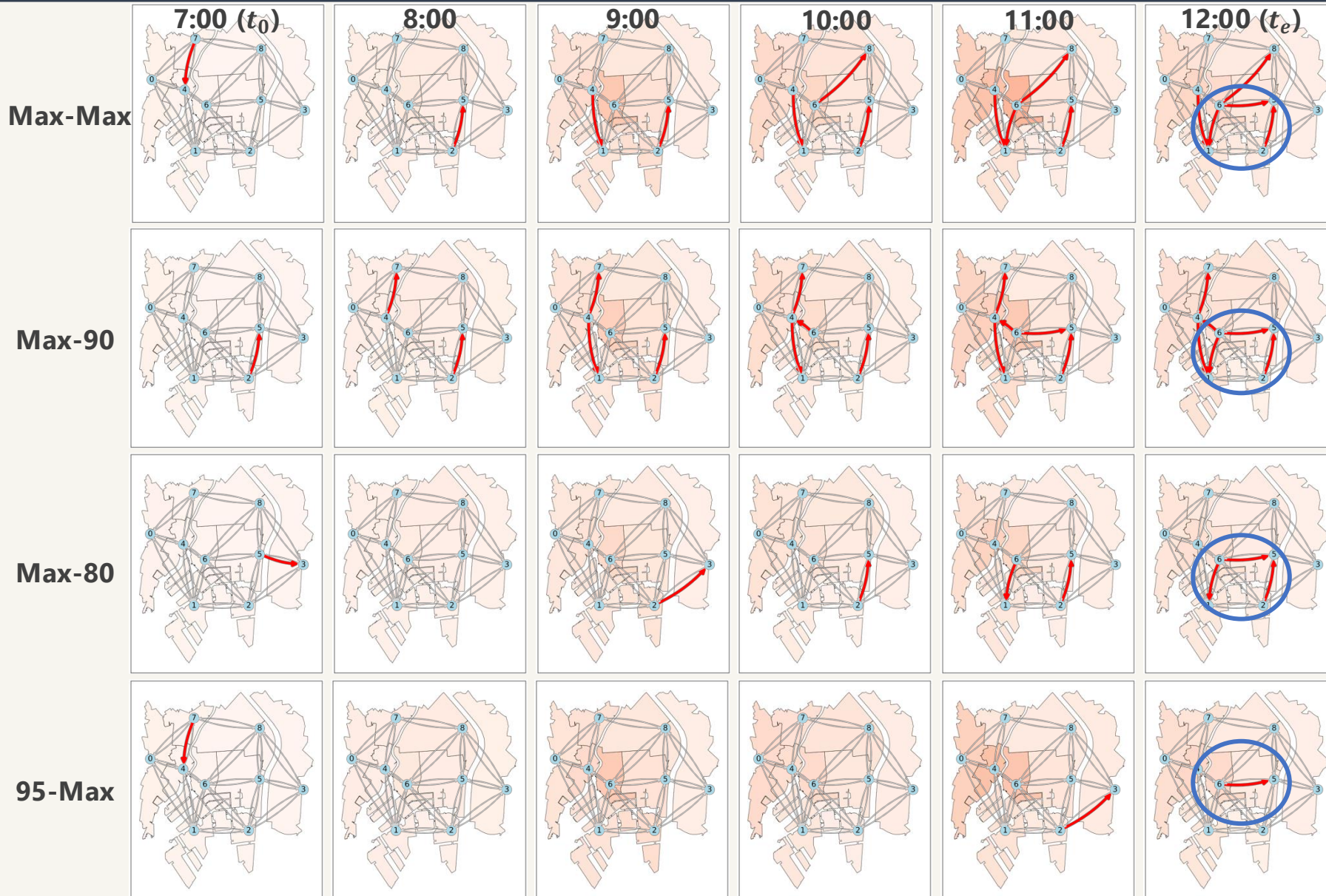
Especially when normal traffic demand is high
(up to 18.1% $\approx$ 17 min.)



Reconfiguration can disrupt traffic, but an optimal transition mitigates the disruption.

Optimal transition sequences

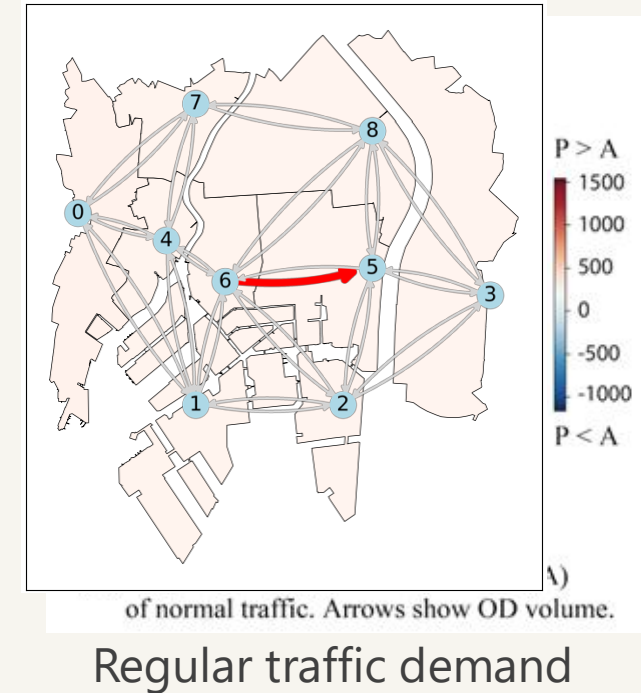
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→ Contraflow link

A robust contraflow pattern across demand scenarios

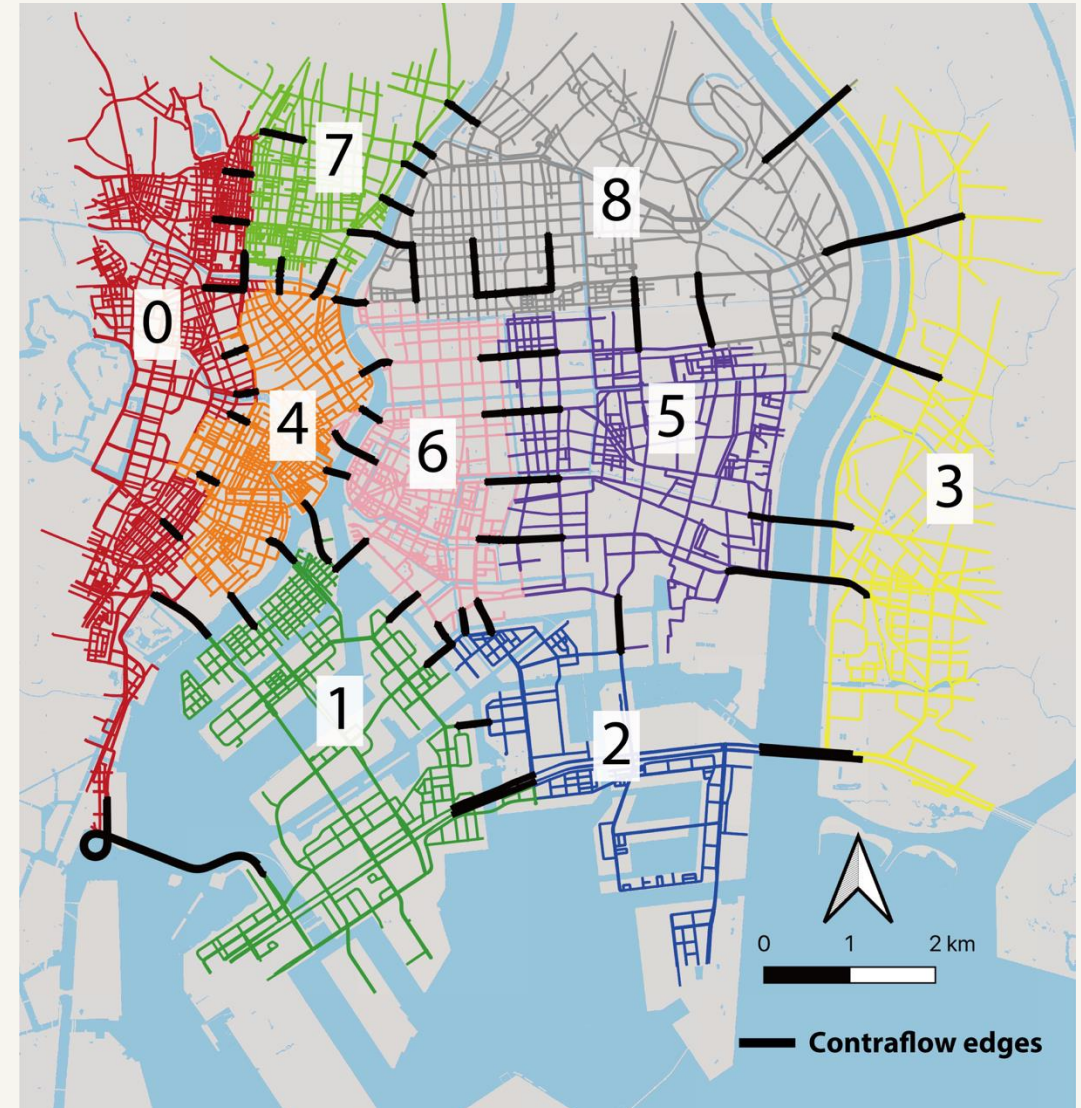
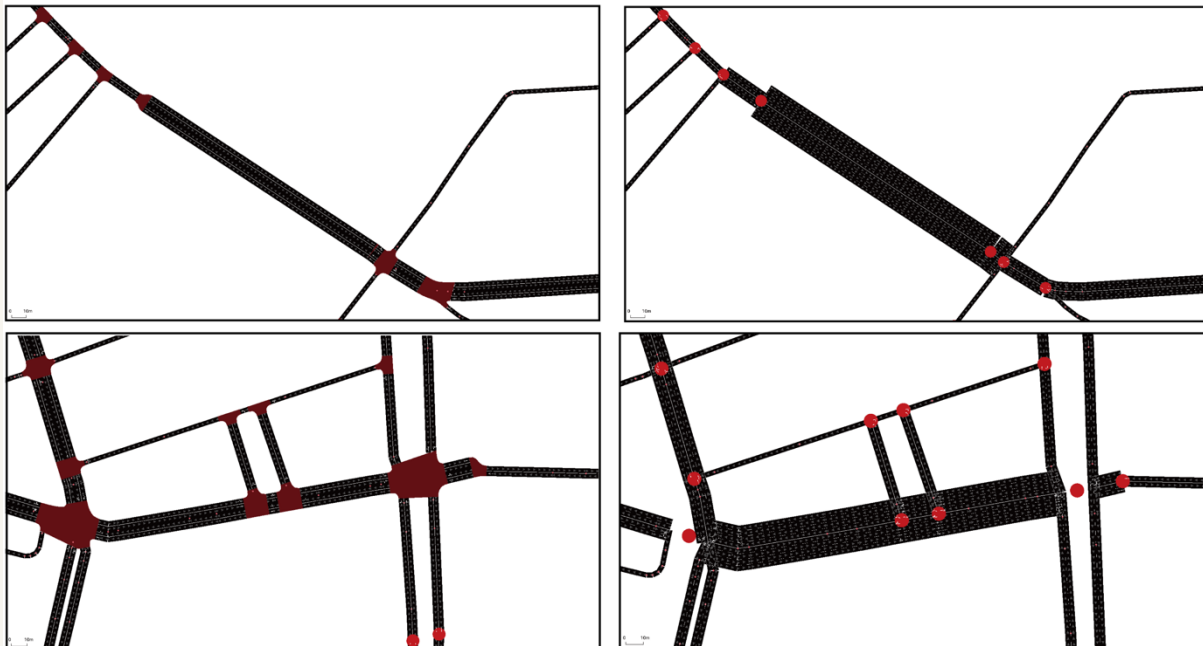
The link 6→5 is activated later.
= Robust transition pattern



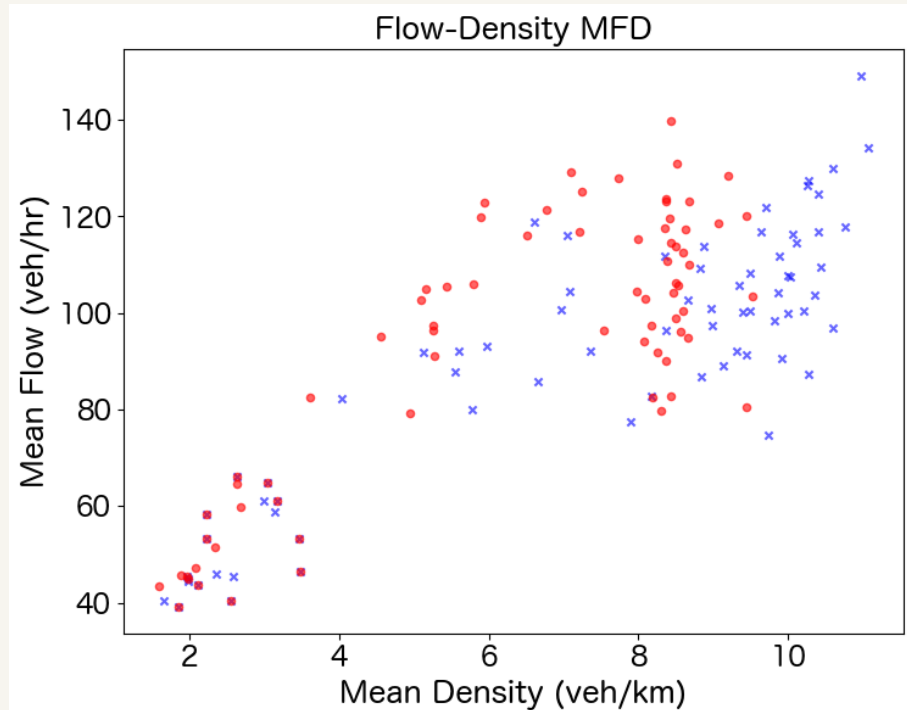
- A microscopic simulator, SUMO, is used to validate the optimization results of the macroscopic simulator.
- 214 links are selected for contraflow.
 - The capacity of the links is **doubled** when control is implemented.
- Demand is a Max-Max scenario.

Without contraflow

With contraflow



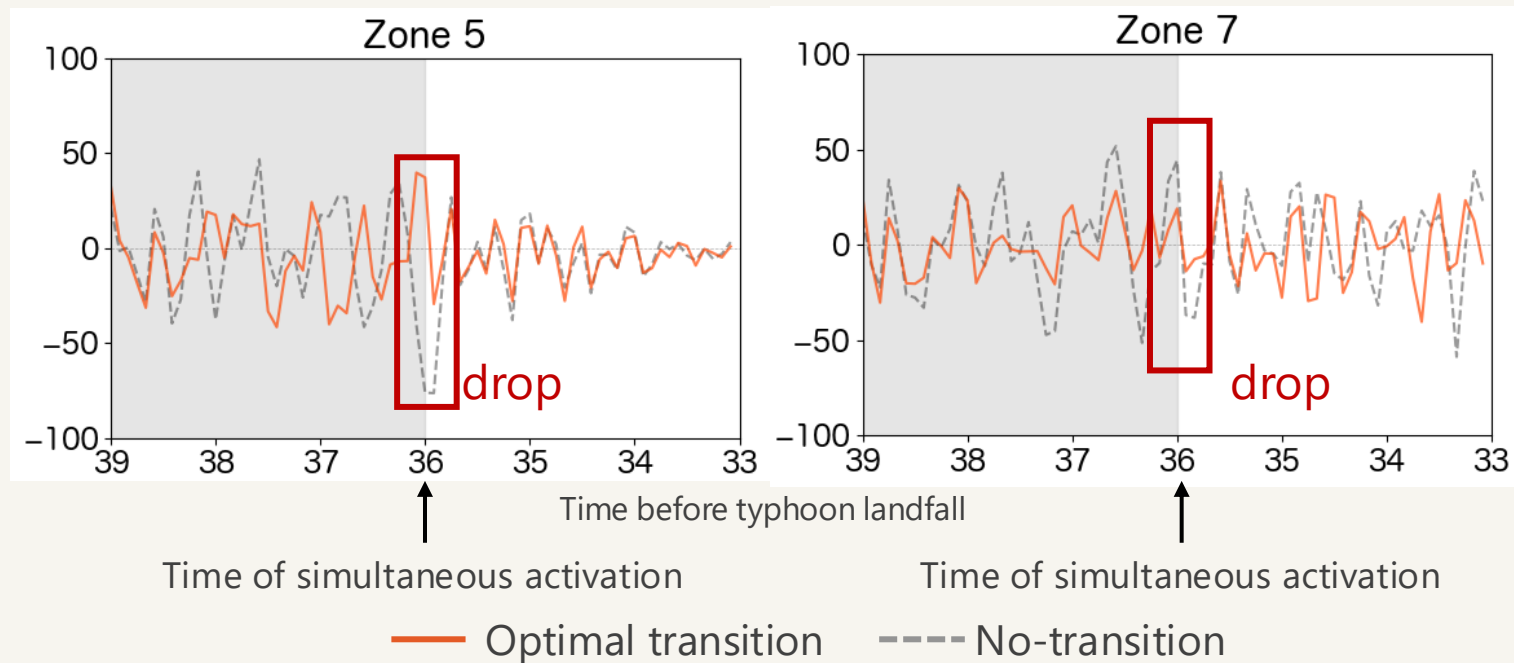
MFD of the whole network



● Optimal transition × Random transition (worst case)

Ensure that the optimal transition outperforms the random transition.

Change rate of link average speed



- The no-transition case shows a large **drop in speed** at the time of contraflow.
→ Flow disruption by the abrupt change
- Reconfiguration **levels the local congestion** by control over time.

The advantage of optimal reconfiguration.

- A flexible and adaptable planning scheme.
- Random transitions can degrade traffic, but an optimal transition minimizes the impact.
- Levels the local congestion by traffic control over the planning horizon.

Macro-micro cooperation.

- MFD-based model successfully found the effective reconfiguration plan.
- Microscopic simulation helps verify the macroscopic plan and evaluate the link-level congestion.

Sampling-space restriction speeds up optimization.

- A large but constrained solution space makes it difficult to sampling-based heuristics.
- A method of sampling-space restriction speeds up computation by using efficient ZDD-based enumeration.

Chapter 4

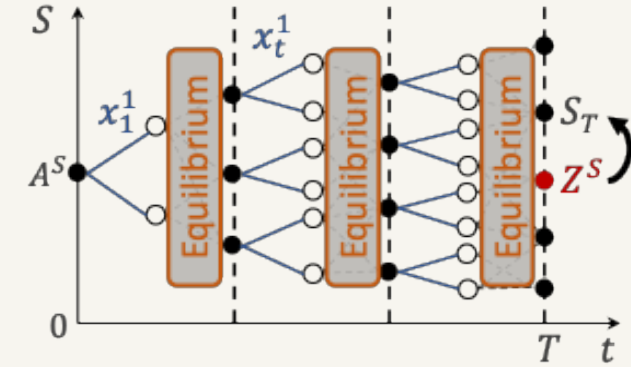
Information design of emergency traffic control

What this overlooks are the general equilibrium effects of information. When drivers with communication devices receive information and alter their behavior, they affect driving conditions for others, both those with devices and those without.

Arnott, Richard, Andre De Palma, and Robin Lindsey.

"Does providing information to drivers reduce traffic congestion?." *Transportation Research Part A: General* 25.5 (1991): 309-318.

Information about traffic control can alter people's behavior, concentrate demand, and worsen congestion (**information paradox**).



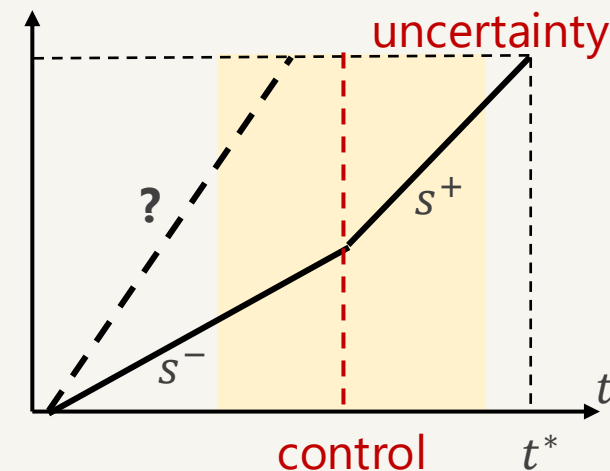
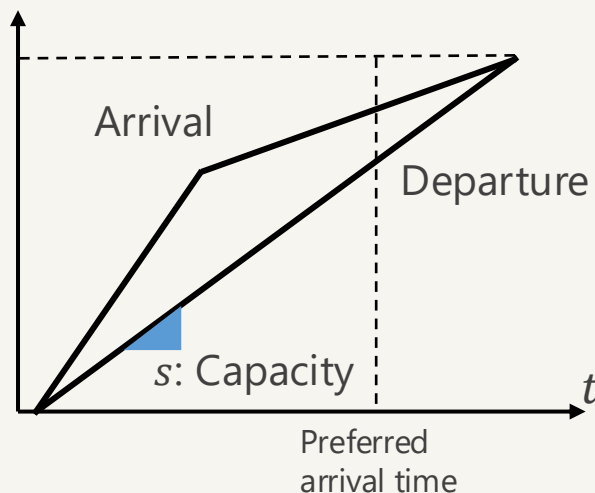
Research question

- How does the amount of information affect users' departure time choices and the system cost?
- How should a traffic operator distribute such information to mitigate traffic congestion?

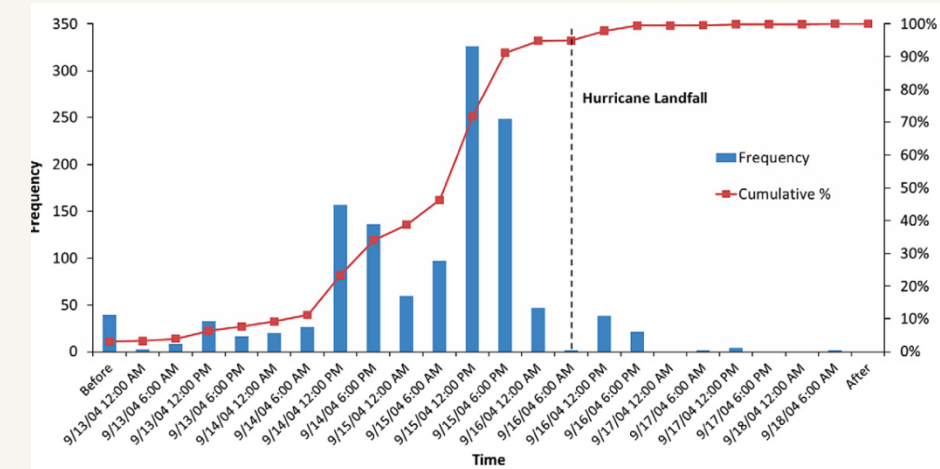
Model

- Equilibrium model of departure time choice under **uncertainty about within-day capacity enhancement**.

Bottleneck model
(Vickrey, 1969)



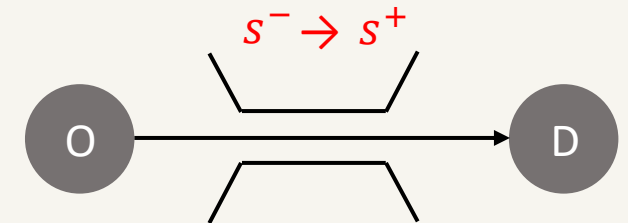
- In reality, evacuation departures are not at equilibrium.
→ **"S-curve"**: rise of departure following evacuation orders.
 - Empirical studies on hurricane evacuation revealed:
 - Those worried about traffic were more likely to depart earlier (Wong, 2018)
 - Pre-departure info on travel delays move departure earlier (Bian, 2022)
- **People respond to congestion and adjust their departure time.**
- Most people complete evacuation by landfall → **Common desired arrival time.**



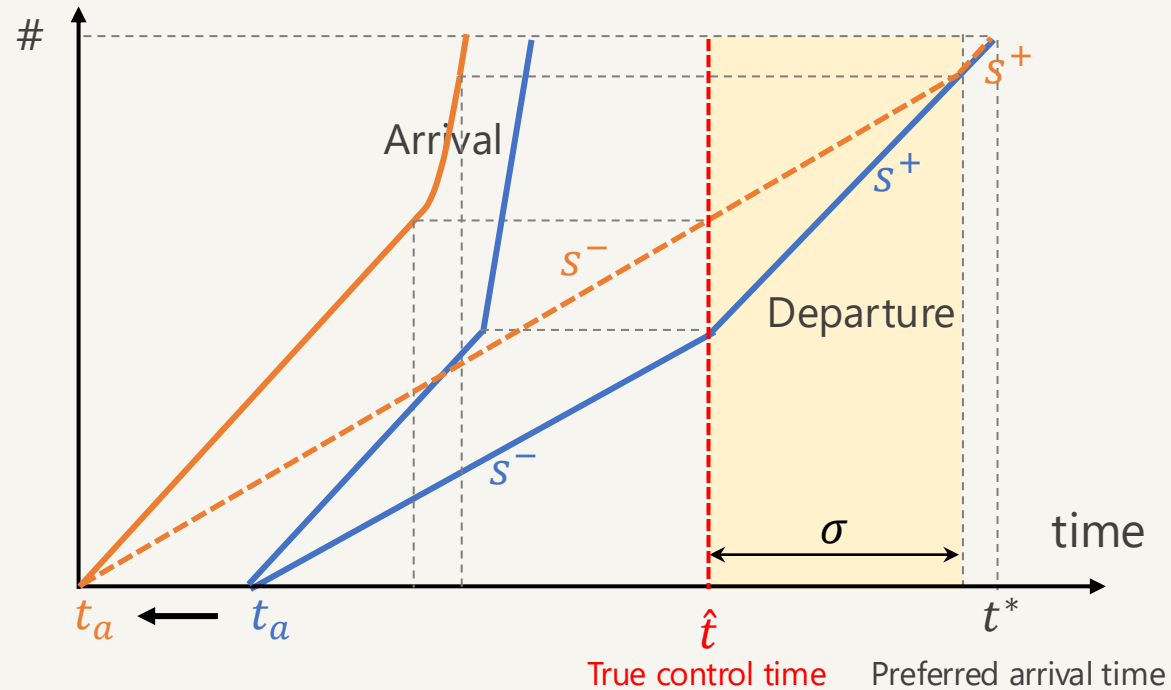
Departure curve of Hurricane Ivan (Hasan, 2013)

- An equilibrium model is used
 - to model the **tradeoff** between congestion avoidance and early arrival.
 - to investigate the **mechanism** of information and departure time adjustment under capacity change.
- Ignore network structure, route choice, preference heterogeneity, and bounded rationality
→ **One-directional, recurrent mass evacuation with information on traffic control.**

- Single OD, single route
- **Exogenous capacity change**
- **Within-day enhancement**
- **Travel cost**
 - Users minimize travel costs associated with **Schedule Delay Early (SDE)** and **queueing time** at the bottleneck.
 - **The cost of late arrival is sufficiently high.**
- **User Equilibrium**
 - made for analytical tractability to capture the aggregate impact of information and control.
- **Partial information**
 - Users know the possible *range of control timing*.
The probability of control activation is uniformly distributed within this range.
- The preferred arrival time is identical for all users.
- Users are risk-neutral, considering only the expected travel cost.
- Free-flow travel time is normalized to zero.



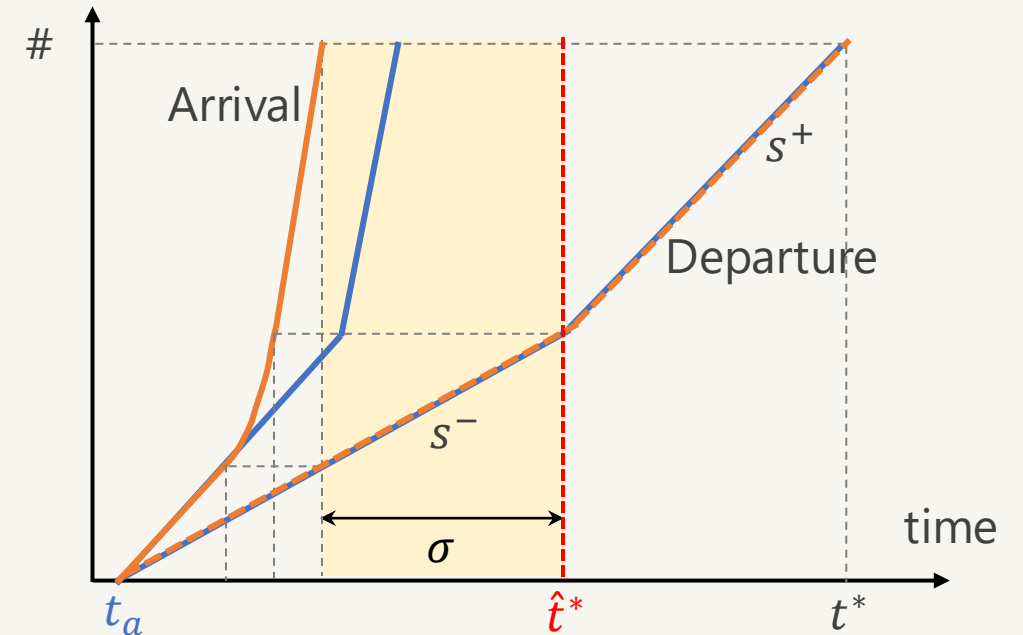
Perfect information and later-biased information



- All users must complete the trip by t^* , even if the control is the latest possible.
- Cumulative arrival $A(t)$ is derived from expected cost minimization.

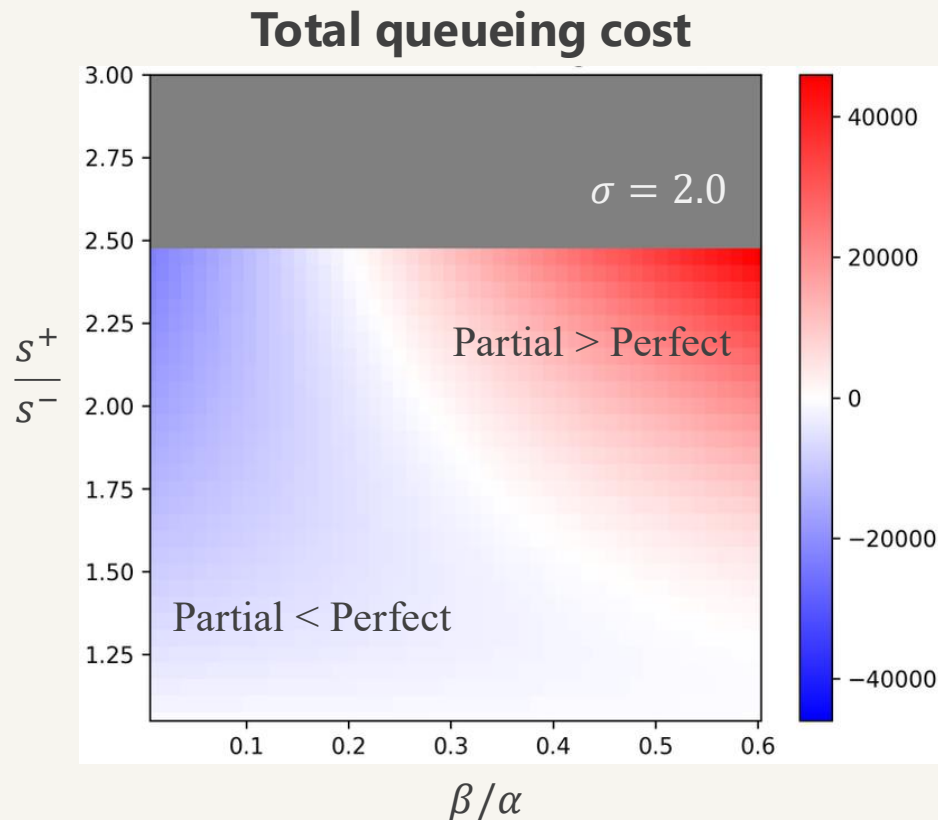
$$\frac{dE}{dt} = 0, C(t) = \mathbb{E}_{\hat{t}}[\alpha T(t, \hat{t}) + \beta \max(0, t^* - t - T(t, \hat{t}))]$$

Perfect information and earlier-biased information



- Later-biased information promotes earlier departure. **Dispersing effect**
- Earlier-biased information always worsen queueing cost.
- In terms of total cost, perfect information is always better than partial information.

- From the operator's perspective, **giving partial information can be better than giving perfect information.**
- The operator seeks to **minimize the total queueing time**.
 - Congestion during evacuation delays evacuation completion and even leads to non-evacuation.
- Later-biased information can lower total queueing time compared to perfect information.

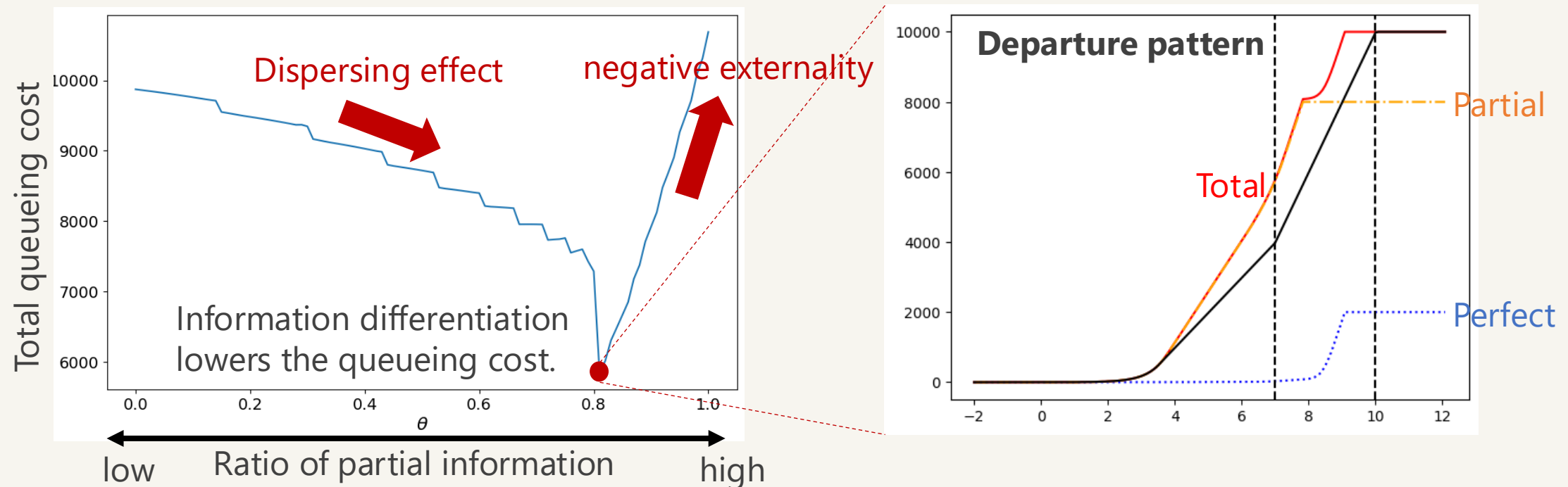


- When β (the marginal cost of schedule delay early) **is small**, later-biased information reduces queueing cost more than perfect information does.
- Users are flexible enough to hedge against the possible late arrival
→ depart earlier and distribute departures.

- The queueing cost is further reduced by using the different departure patterns.
→ Disperse concentration by giving users heterogeneous information

Equilibrium under heterogeneous information

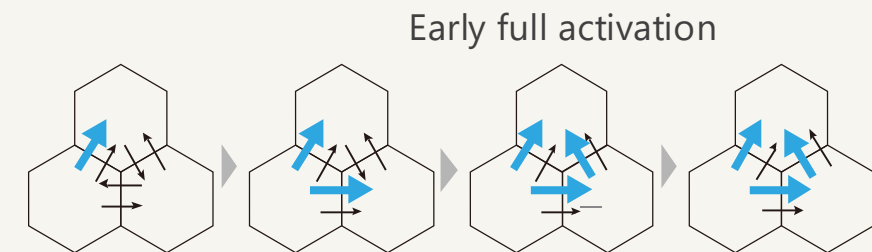
- Only a subset of users has partial information: others have perfect information.
- Each user group responds to the *aggregate* arrival rate $r(t) = r_{\text{perfect}}(t) + r_{\text{partial}}(t)$ (**mean-field game**)
- Nash equilibrium is satisfied within each user group. $r_k(t) > 0 \Rightarrow \bar{C}_k(t; r) = \inf_{\tau \in \mathcal{T}} \bar{C}_k(\tau; r), \forall k \in \{\text{perfect}, \text{partial}\}$



1. When the cost of early arrival is low, **announce a later time window.**
2. When travelers insist on the desired arrival time, **announce the exact control time.**
3. **Lower cost of early arrival.**
 - Spread out departure times and reduce both total cost and queueing time.
 - e.g., incentives for early departure, improving destination environment, and raising risk awareness (Jiang et al., 2022)
4. **Provide later-biased information in high-risk areas and perfect information in low-risk areas.**
 - Make the departure of high-risk areas earlier, and low-risk areas later.
 - Prevent shadow evacuation

Insights into reconfiguration planning

- Reconfiguration planning = earlier-biased information.
- Earlier-biased information is always worse than perfect information.
 - The operator should **commit** to the timing when the control is fully activated (perfect information).



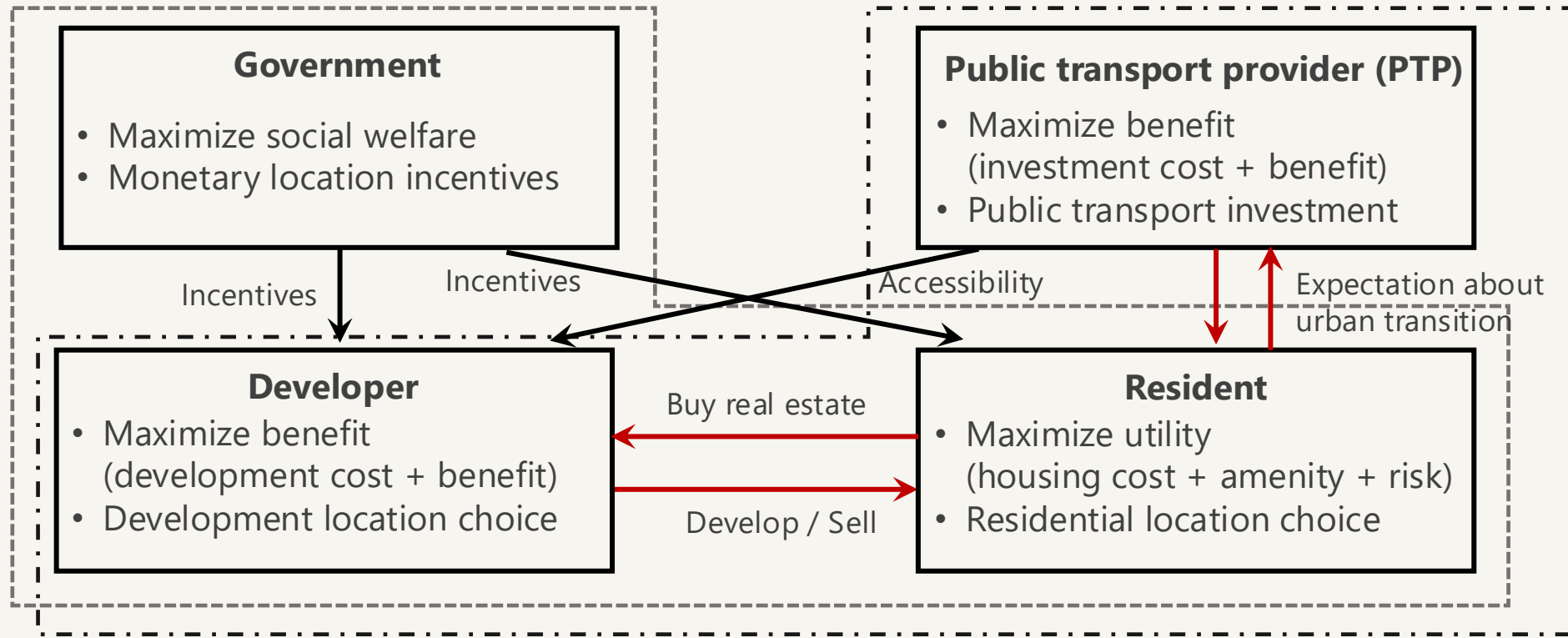
Chapter 5

Dynamic game-theoretic model of relocation, development, and transport investment in disaster-prone areas

However, in dynamic models where current unobservables are not necessarily independent of past history, or in "thin" markets where there is strategic interaction between agents, feedbacks become strong enough so that it is necessary to deal with endogeneity in explanatory variables.

McFadden, Daniel.

"Economic choices." *American economic review*, 91.3 (2001): 351-378.



① Extensive-form game

- Interaction between developers and residents
- Optimal **location incentives** of government
- Model-based analysis

② Markov game

- Interaction between developers and PTP
- **Public transit-oriented land-use transition**
- Empirical data of developers, PTP, and residents
- Estimation of the strategic interaction

Quantitative Urban/Spatial Economics (Redding & Rossi-Hansberg, 2017)

- A general-equilibrium framework that links spatial theory to rich geographic data.
- Enables structurally interpretable estimation and policy counterfactuals across many locations.
- Mainly static

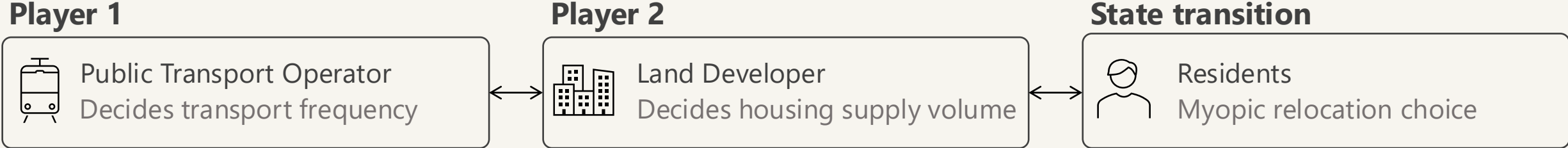
Why dynamic in disasters?

- **Forward-looking:** protection, infrastructure, and relocation are forward-looking decisions under future risks.
- **Transition paths matter:** Policies and investments reshape long-run population dynamics, not just the short-run effect.

→ Motivates the literature on dynamic migration and spatial economic activity.

	Spatial scale	Forward-looking players	Other players	Treatment of dynamic optimization
Sun (2025)	National	Households	Developers	Solve DP
Takeda & Yamagishi (2025)	City	Households	Firms, Developers	Observed dynamics as option value
Hsiao (2023)	City	Developers, Government	Households	Prices as option values
This thesis	City	Developers, Transport operator	Households	Markov game + Monte-Carlo sampling for value function approximation

► Estimate the magnitude and direction of strategic interactions from observed data.

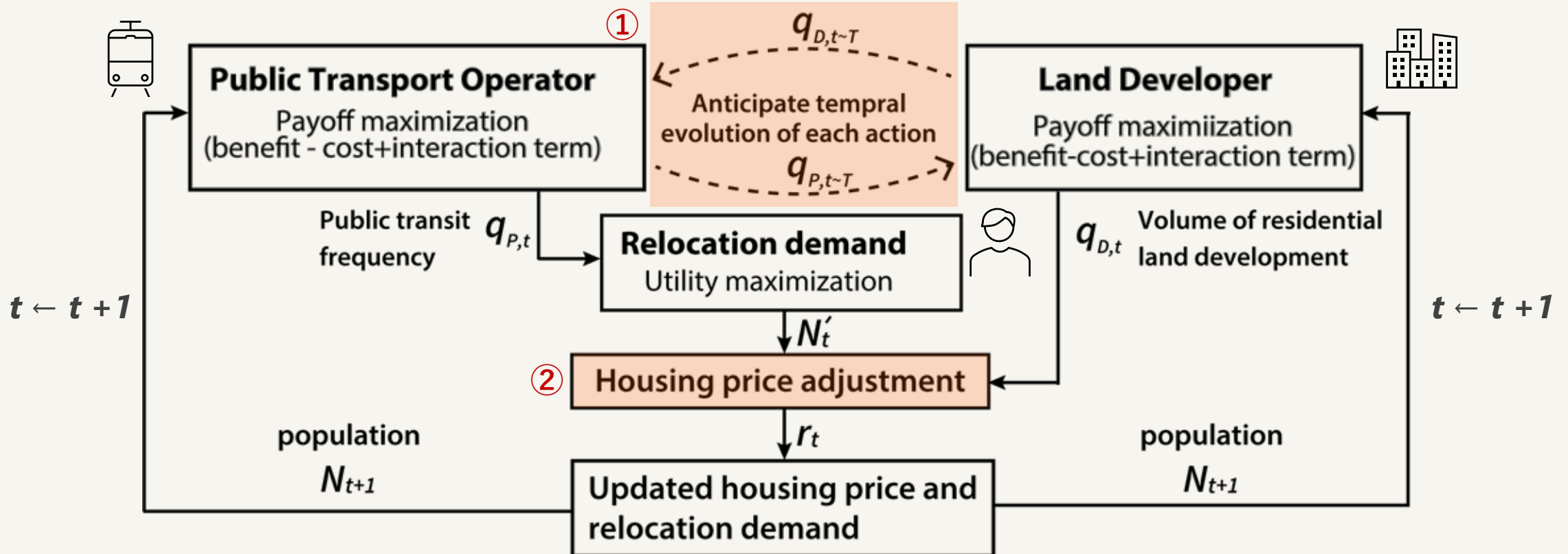


① Markov Perfect Equilibrium (MPE)

Each player maximizes expected present value given others' strategies.

② Demand-Supply Equilibrium

Housing demand and supply coincide in every period.



- 
- port service level.

Prob.

1

$\tilde{F}_i(q_i | S)$

$G(v)$

0

action

q

q'

$$\Pr(\sigma_i(S, v_i) \leq q_i) = \tilde{F}_i(q_i|S)$$

Observed distribution of q_i under S

Probabilistic action under state S_t and shock v

► $\sigma_i(S_t, v_{it}; \theta) = \tilde{F}_i^{-1}(G_i(v_i | S_t) | S_t)$

- Each player i maximizes the discounted stream of expected payoffs under the equilibrium policy σ

$$\max V_i(S_t; \sigma) = \mathbb{E} \left[\underbrace{\pi_i(\sigma_t, S_t, \nu_{i,t})}_{\text{instantaneous payoff}} + \underbrace{\beta_i}_{\text{discount factor}} \int \underbrace{V_i(S_{t+1}) dP(S_{t+1} | S_t, \sigma_t)}_{\text{State transition probability}} \right]$$

= Residents' relocation choice model

- **Markov Perfect Equilibrium (MPE)**

$$V_i(S_t; \sigma) \geq V_i(S_t; \sigma'_i, \sigma_{-i}) \quad \forall S_t, v_{i,t}, \sigma'_i \in \Sigma_i,$$

A strategy profile $\sigma = (\sigma_p, \sigma_D)$ is an MPE if no unilateral deviation improves expected value.

► Estimation is conducted to satisfy the constraint

- **Demand-supply equilibrium**

$$\rho M_t = q_{D,t},$$

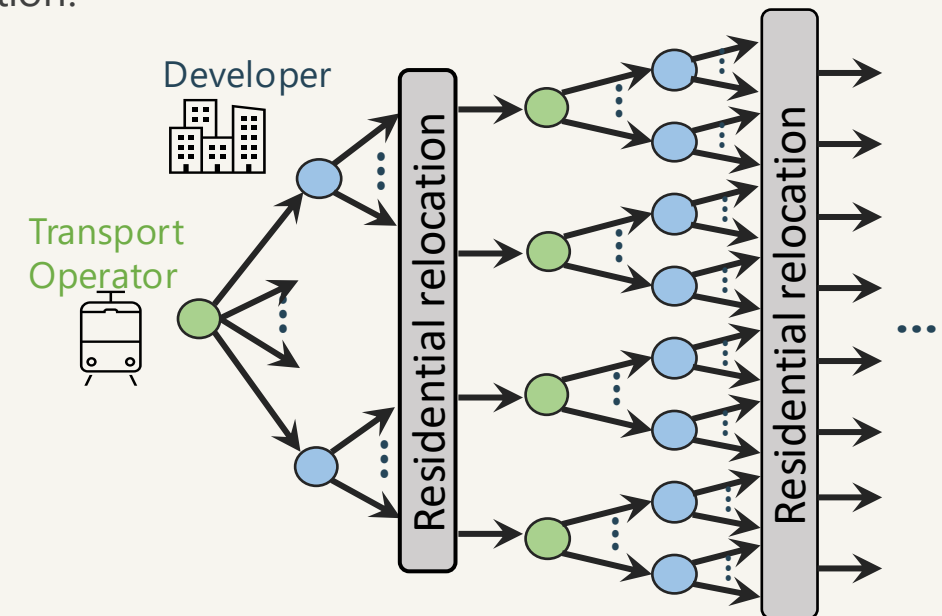
Housing demand Housing supply
by residents' behavior by developer's behavior

► Convergence calculation is conducted in each time step in the estimation.

Next step ➡

Estimate payoff parameters θ of each player in payoff π so that both MPE and demand-supply equilibrium is satisfied.

Continuous actions and high-dimensional state space
→ **Hard to calculate value function of all states.**



Two-step estimation by Bajari et al. (2007)

Stage 1

Estimate Policy Functions & State Transitions

- Estimate policy functions $\sigma_i(S_t)$ directly from observed data $\{S_t, q_t\}$.
- Estimate state transition probabilities $P(S_{t+1} | S_t, q_t)$.

Residential location choice model:

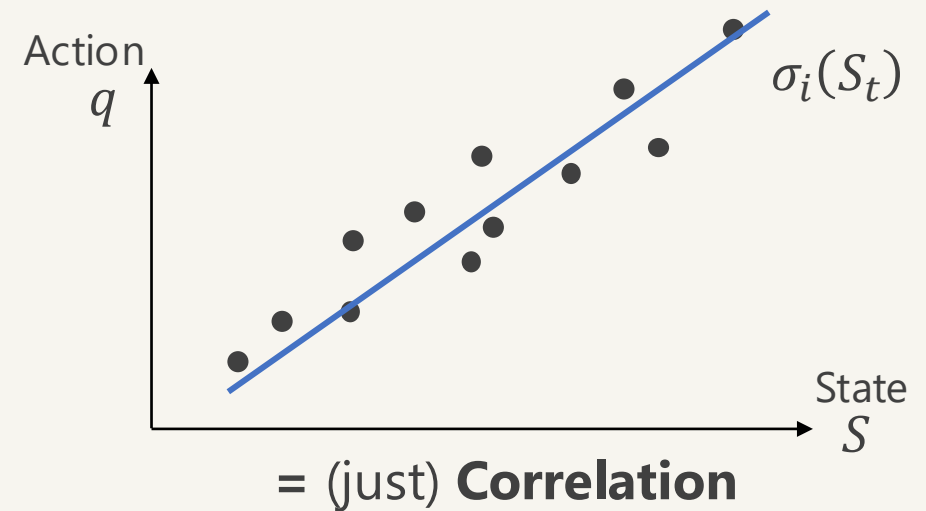
Mixed logit model to incorporate spatial correlation.



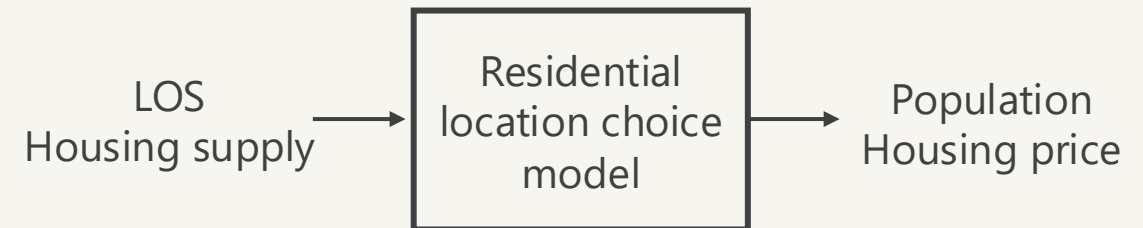
Stage 2

Estimate Payoff Parameters

Policy function: "**What** the players do"



State transition: "**How** the environment reacts"



Two-step estimation by Bajari et al. (2007)

Stage 1

Estimate Policy Functions & State Transitions



Stage 2

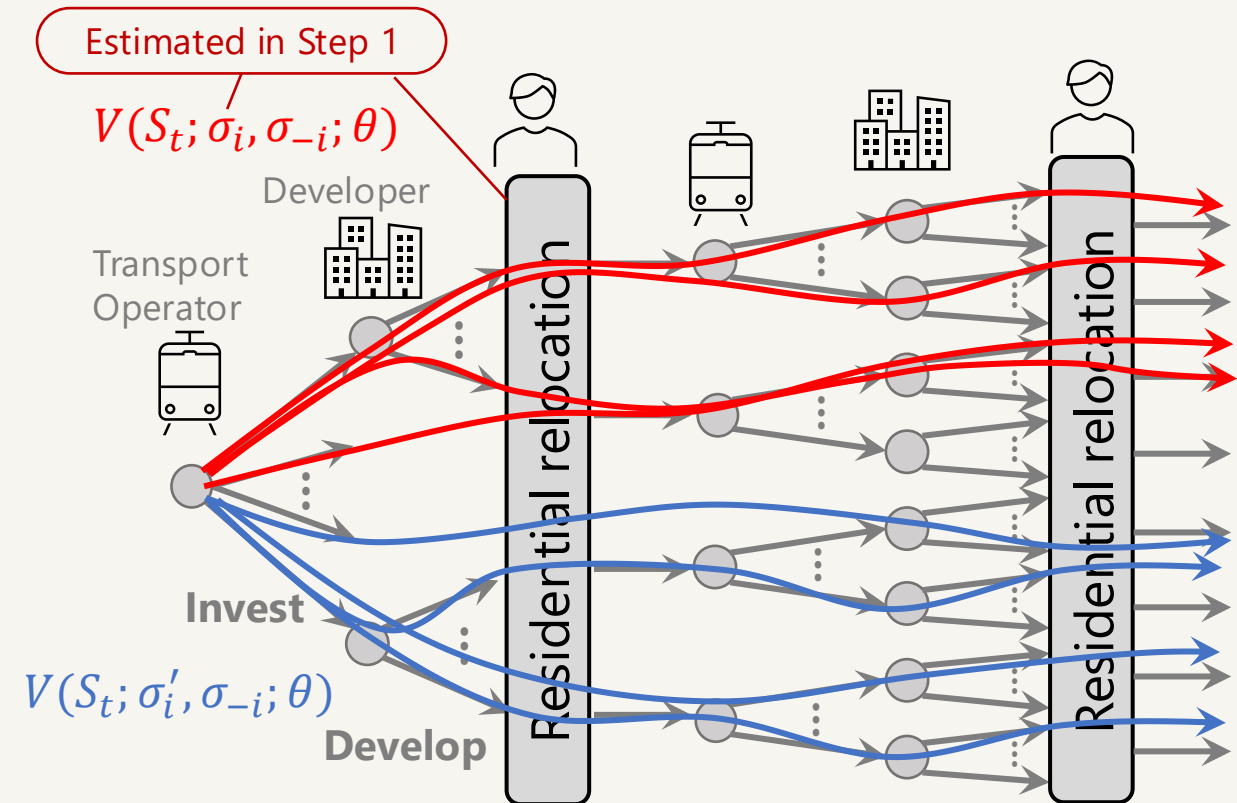
Estimate Payoff Parameters

- Use **Monte Carlo sampling** to approximate value functions V_i .
- Minimize loss function $Q(\theta)$ based on MPE optimality conditions.

$$Q(\theta) = \sum_i \sum_t \sum_{a'_i \neq a_{i,t}} \min\{V(S_t; \sigma_i, \sigma_{-i}; \theta_i) - V(S_t; \sigma'_i, \sigma_{-i}; \theta_i), 0\}^2$$

= Make the observed actions optimal.

"Why the players behave as they do"



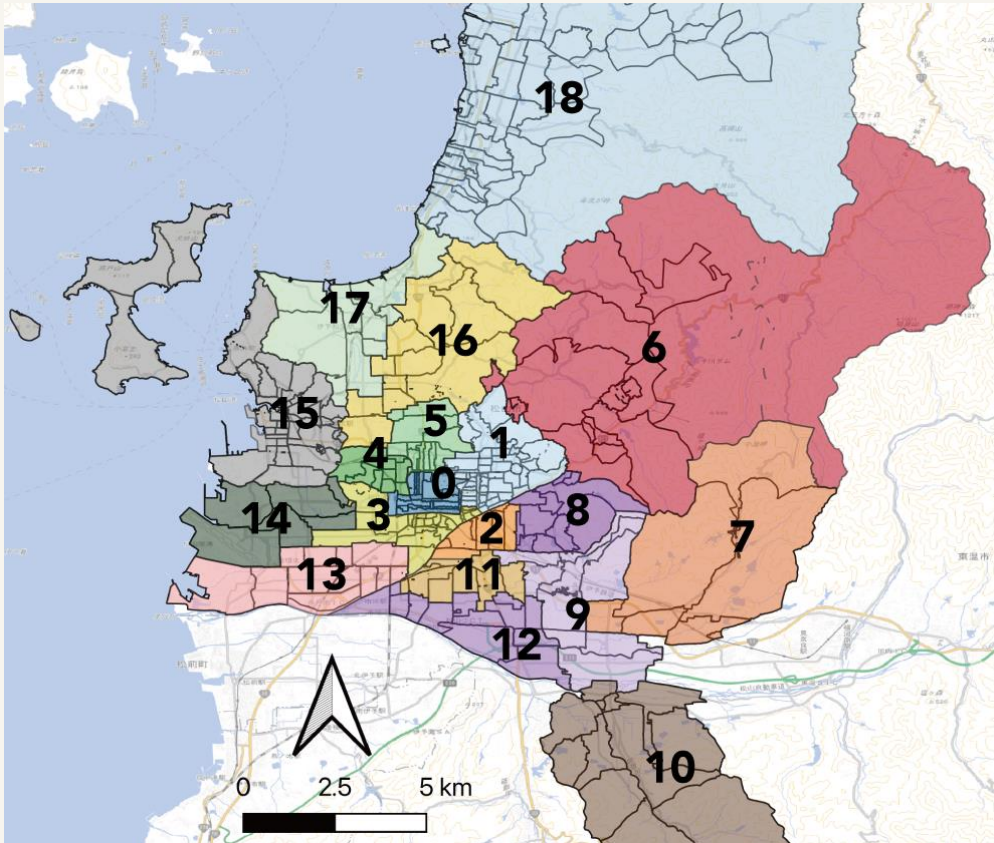
If the observed data follows MPE, θ should satisfy

$$V(S_t; \sigma_i, \sigma_{-i}; \theta) \geq V(S_t; \sigma'_i, \sigma_{-i}; \theta)$$

Zoning

Residential relocation: 39 zones

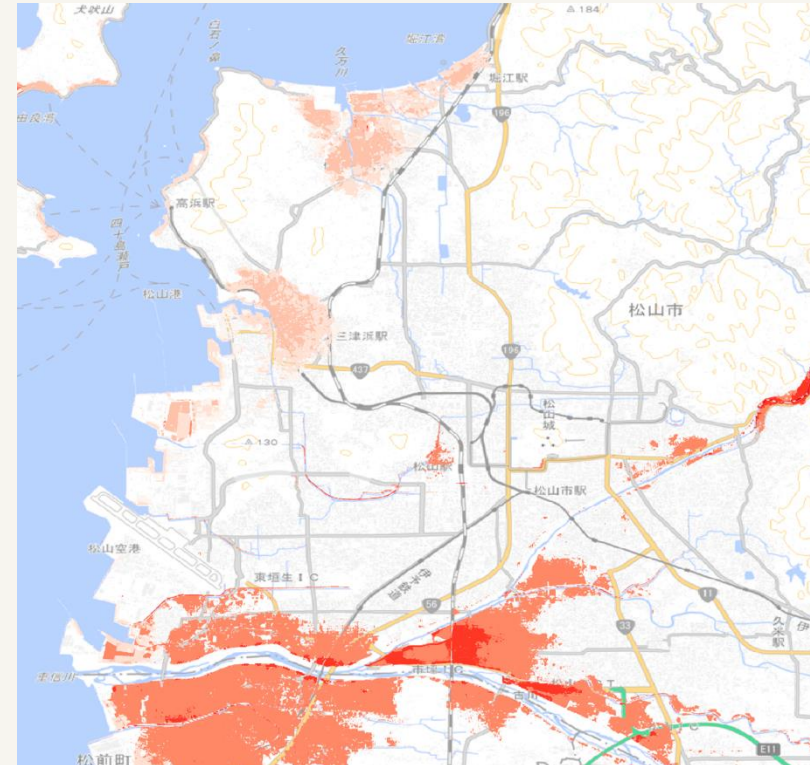
Develop/invest: 19 zones



Context: Disaster risk

Flood: Inundation areas of 1/100 floods used.

Tsunami: Expected inundation areas of large earthquake.



Urbanization in high-risk areas.

→ Need for effective land-use control

Supply-side



Public Transport

Lines and timetables of buses, trams, and railways (2008-2023)



Land development

Building permits of all the buildings developed between 2008 and 2023.
35,239 records

Demand-side



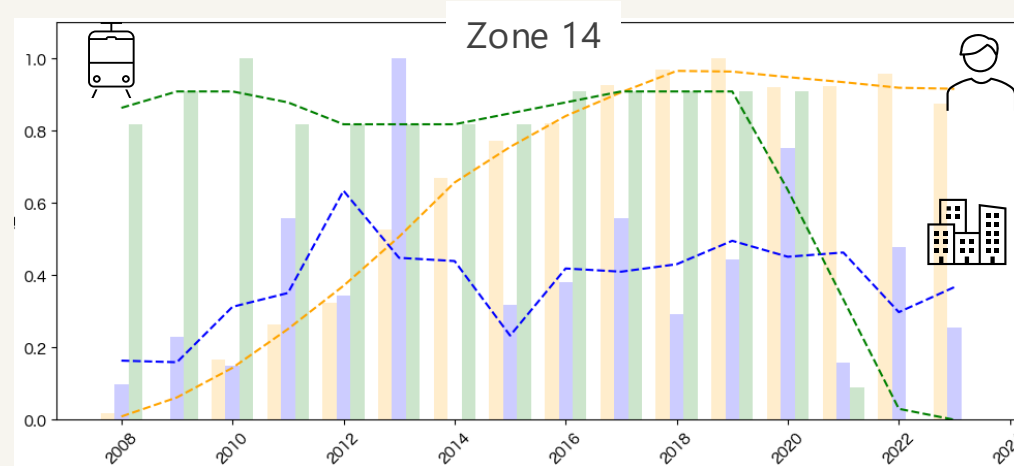
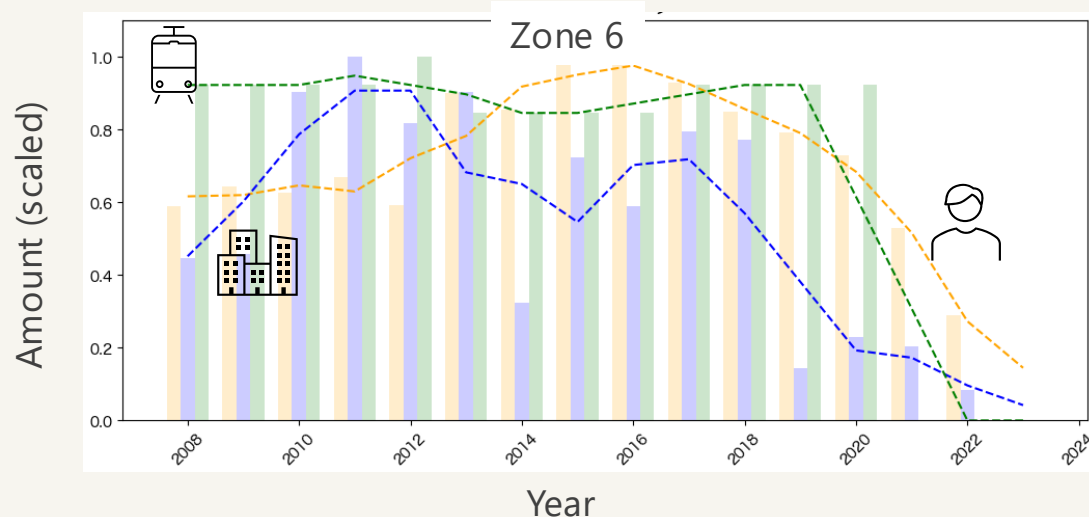
Relocation history

Survey result on current and previous house location.
635 records.



Aggregate migration

Registered population and aggregated flow between districts.



Stage 1: Estimates of residential relocation model (Mixed Logit model)

43

		Baseline	Age 65+	Car owner	With children	Move-in
Dist. home		-6.151 (-22.51)				
Dist. CBD		0.146 (0.58)				-3.647 (-7.44)
PT freq.		0.083 (1.85)		-0.115 (-3.02)		
Risk area		0.327 (4.34)				
Dist. school		-0.143 (-1.39)			-0.512 (-2.44)	
Commercial land		-0.074 (-0.76)	-0.188 (-1.88)		-0.204 (-2.94)	
Housing price		-0.485 (-2.98)				
Constant		1.431 (4.33)				
Spatial correlation parameter ρ		3.322 (81.83)				
Observations		351	832			
Adjusted R ²		0.063				
initial log-likelihood			2667.9			
final log-likelihood			2187.9			
Mcfadden's adjusted R ²			0.182			

Strong preference for **nearby zones**.

City center is preferred by people moved in from outside.

Those who do not have cars care much about public transport.

Risk areas are favored even after adjusting housing prices.



Utility of increasing PT frequencies by **58 [runs/day]**
= Utility of decreasing housing price by **10,000 [JPY/m²]**,
especially for non-car owners.



Population

Transit LOS

Distance from
CBD

Development
volume

$$\pi_P = pop \cdot q_P^\eta - (\theta_{P,c} + \theta_{P,c \cdot d} dist) \cdot q_P + \theta_{P,D} \cdot q_D$$

Revenue

Cost

Strategic
interaction



$$\pi_D = price \cdot pop - (\theta_{D,c} + \theta_{D,c \cdot d} dist) \cdot q_D + \theta_{D,P} \cdot q_P$$

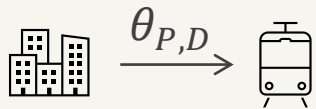
Land price

Population

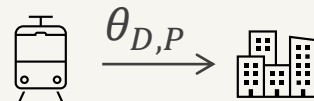
Development
volume

Transit LOS

		Parameter	Estimate	Std. Error	95% Confidence Interval
	cost	$\theta_{P,c}$	0.494	0.13	(0.242, 0.783)
	distance effect	$\theta_{P,c \cdot d}$	0.260	3.69	(-7.728, 4.246)
	interaction  $\rightarrow$ 	$\theta_{P,D}$	-0.274	1.96	(-3.561, 3.774)
	cost	$\theta_{D,c}$	6.915	1.59	(5.168, 10.572)
	distance effect	$\theta_{D,c \cdot d}$	-0.642	1.06	(-2.066, 1.670)
	interaction  $\rightarrow$ 	$\theta_{D,P}$	<u>19.988</u>	5.75	(13.568, 32.405)



Increasing housing development has no statistically significant effect on public transport provision.



The investment in public transport affects the housing development **positively**.

- Increasing PT frequencies by 10 [runs/day] in a zone is equivalent to 2 million [JPY/year] for developers.



Potential of public transit-oriented land-use control.

1. Subsidizing public transport in low-risk areas.

(Service levels are increased by 100 runs/day)

Overall welfare changes [%]

	Subsidized zone		
	Zone 2	Zone 3	Zone 14
Transport operator	3.31	3.31	12.86
Developer	9.93	9.93	55.39
Resident (with car)	0.046	0.049	0.016
Resident (without car)	0.059	0.075	0.029

2. Housing development regulations in high-risk areas.

(Supply is limited to half of the 2008 level.)

Overall welfare changes [%]

	Regulated zone		
	Zone 11	Zone 12	Zone 13
Transport operator	12.14	12.14	47.82
Developer	35.50	35.50	74.60
Resident (with car)	-0.11	-0.004	-0.31
Resident (without car)	-0.10	-0.07	-0.29

- Welfare **gain** of residents

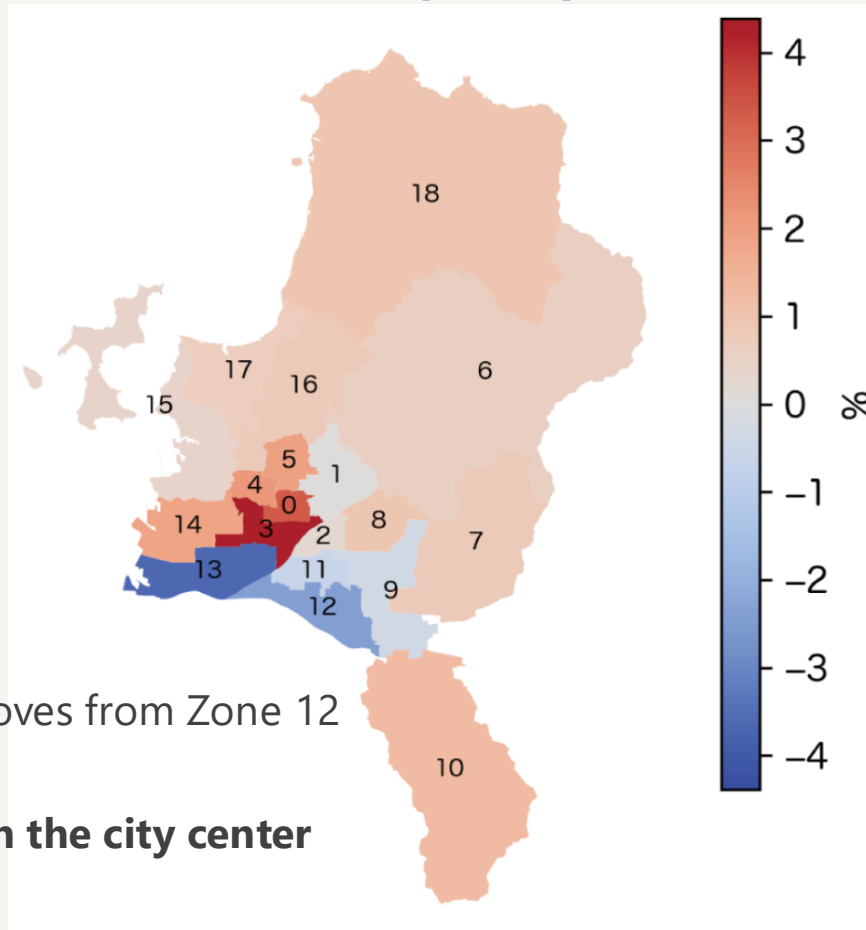
- Especially for residents without cars
- Due to increased public transport supply + increased housing supply (positive feedback)

- Welfare **loss** of residents

by increased housing prices due to supply reduction.

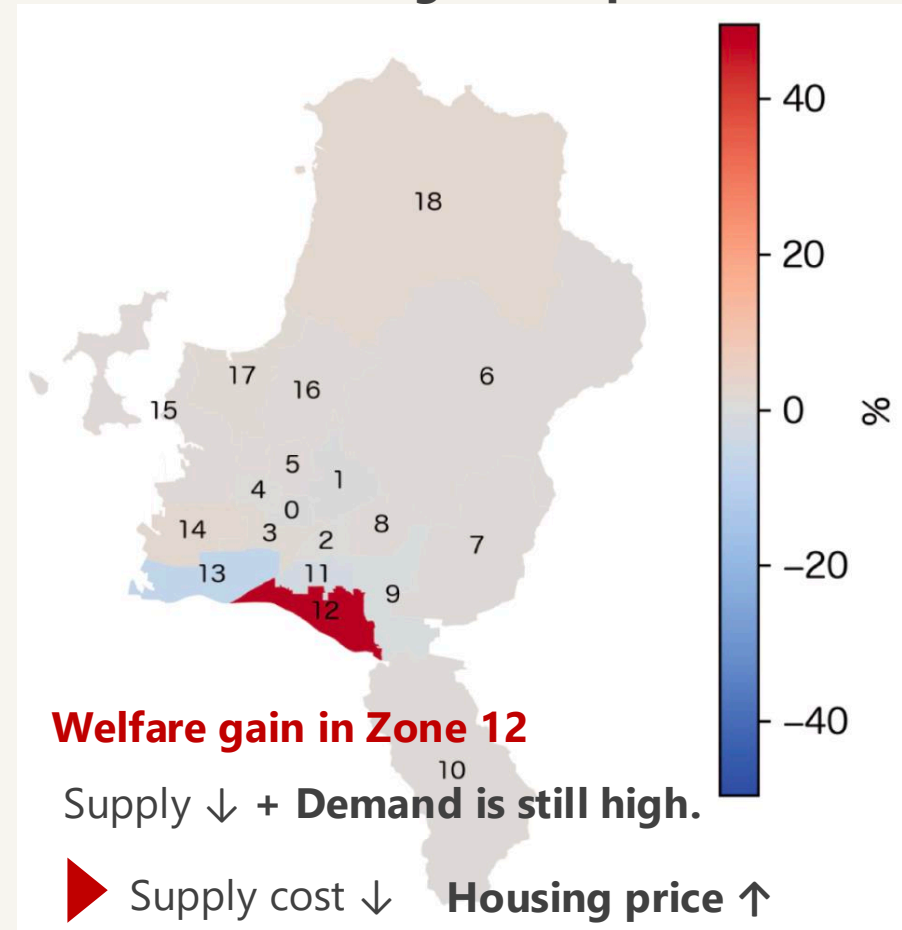
Welfare change by development regulation in high-risk areas (Zone 12)

Public transport operator



The population moves from Zone 12 to the city center
→ **Higher profit in the city center**

Housing developer



Strategic interaction between public transport providers and developers.

- Incorporating dynamic expectations in a land-use transport interaction model.
- Efficient estimation procedure for empirical data of public transit, development, and residential relocation.
- Asymmetric interaction between the two players.

Policies with the same objective of risk reduction lead to different welfare changes.

- Transit investment increased residents' welfare, attracting migrants, more housing development, and a decrease in housing prices (feedback effect)
- Importance of evaluating the reconfiguration process in light of **endogenous feedback effects**.

Panel data on both supply- and demand-side behaviors enable the estimation of the feedback effect.

- Collecting and combining multidimensional data enhance understandings of system dynamics.

1. Reconfiguration process design for staged and flexible control

Chap. 5

Staged evacuation traffic control

Chap. 3

- Proposed staged reconfiguration framework from normal to disaster time control.
- Developed **a method of sampling-space restriction using ZDD.**
- The effectiveness of reconfiguration is confirmed at both macro- and micro-scales.

Urban transition toward resilient land-use patterns

- Modeled **the dynamic change** of population distribution through **interactions with land development and public transport provision.**
= Multi-player, multi-period model

2. Understanding dynamic interactions in reconfiguration processes

Information design of traffic control

Chap. 4

- Modeled **the evacuation departure time equilibrium under uncertain within-day traffic control**
- Uncertain information about traffic control moves evacuees' expectations, alleviating congestion.

Short-term

Strategic interactions of location behaviors

Chap. 5

- Land/transport supply decisions influenced by expectations about others' locations were modeled as a **dynamic game framework.**
- An efficient estimation method and positive interactions were presented.

Long-term

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